

Finite-Sample Performance of High-Dimensional IVQR Estimators

Supervisor discussion report based on 500 Monte Carlo replications per design

Master Thesis Project

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Executive Summary

This report evaluates whether three instrumental-variable quantile-regression (IVQR) estimators are sufficiently reliable for the thesis: a feasible DML-style IVQR estimator, an infeasible Oracle IVQR benchmark, and a feasible Post-selection IVQR estimator. All numerical evidence comes from the validated $R = 500$ full-run outputs in `results/report_full_run/`. Historical $R = 10$, $R = 20$, and $R = 100$ calibration experiments are intentionally excluded from the evidentiary base.

The validated full run contains 500 replications in each of 144 design cells. DML-IVQR attains **94.90%** coverage on resolved replications, compared with **94.19%** for Oracle IVQR and **92.42%** for Post-selection IVQR. DML's stronger calibration comes with a mean confidence-region length of 2.959 and a 33.09% full-grid rate. The central concern is Post-selection: its coverage falls to **90.42%** at $\tau = 0.75$, and the worst reported cell covers only **83.8%**.

The provisional assessment is:

- **DML-IVQR – provisionally acceptable as the main feasible estimator.** Its aggregate coverage is closest to the nominal 95% target and is stable across quantiles. The cost is weak informativeness in difficult designs: its regions are widest and frequently accept the full parameter grid.
- **Oracle IVQR – acceptable as an infeasible benchmark.** It has the best point-estimation performance and useful confidence-region length, but knowing the true support cannot manufacture identifying variation. Its coverage is mildly below nominal overall and weaker in some cells.
- **Post-selection IVQR – not yet validated as a reliable 95% procedure.** It produces Oracle-like region lengths but systematically undercovers, particularly at the upper quantile. It remains informative as a comparison estimator, a negative result, and evidence on the cost of imperfect selection.

These classifications are deliberately provisional. The report supports a supervisor decision; it does not declare final acceptance.

Questions for discussion

1. Is DML-IVQR sufficiently informative for the thesis despite its wide and full-grid confidence regions?
2. Should Post-selection remain as a documented comparison, or should it be redesigned before inclusion?
3. Should point estimation be refined beyond the initial 0.2-spaced alpha grid?

1 Research Objective

The target is the structural quantile treatment coefficient $\alpha_0(\tau)$ in a nonseparable outcome model. At quantile τ , the structural restriction can be written schematically as

$$P\{Y \leq D\alpha_0(\tau) + g_\tau(X) \mid X, Z\} = \tau,$$

where D is endogenous, Z is an excluded instrument, and X is a high-dimensional vector of controls. Ordinary quantile regression conditions on the endogenous treatment and therefore does not generally recover the causal structural quantile effect. IVQR instead uses instrument exogeneity and relevance to identify quantile-specific causal effects (Chernozhukov and Hansen 2005, 2006).

High-dimensional controls create a nuisance-estimation problem. Directly fitting hundreds of controls can be unstable; selecting a sparse subset can introduce selection bias; and flexible nuisance learning may overfit. The simulation therefore compares three roles:

1. **Oracle IVQR** knows the true active support and isolates the cost of nuisance estimation and selection, but is infeasible outside simulation.
2. **Post-selection IVQR** selects controls from the data and then applies IVQR, providing a feasible sparse benchmark.
3. **DML-IVQR** uses cross-fitted nuisance estimation and residualized scores to reduce first-order sensitivity to nuisance-learning error.

Good finite-sample behavior is multidimensional. The primary inferential criterion is empirical coverage near the nominal 95%, but a procedure that always reports the entire grid can cover without being informative. Assessment therefore also requires low bias and RMSE, reasonably short confidence regions, few unresolved replications, and robustness across τ , instrument strength, sample size, dimension, and DGP. No single scalar ranking is sufficient.

2 Confidence Intervals, Confidence Regions, and Coverage

2.1 Point estimates and set estimates

A point estimate $\hat{\alpha}$ selects one candidate value. A confidence set reports all candidate values not rejected by an inferential procedure. In a regular, strongly identified problem the set is often a connected interval. Here the test is inverted over an alpha grid, and the accepted set can be disconnected. **Confidence region** is therefore the default term: calling every set an interval would misdescribe unions of separated accepted components.

The nominal 95% level is a property of the procedure under repeated sampling. Empirical Monte Carlo coverage is the observed fraction of simulation replications in which the reported region contains the known true value. For a group of simulation rows $\mathcal{R}$, the report uses

$$\widehat{\text{Cov}} = \frac{\sum_{r \in \mathcal{R}} 1\{\text{CR}_r \text{ resolved}\} 1\{\alpha_{0,r} \in \text{CR}_r\}}{\sum_{r \in \mathcal{R}} 1\{\text{CR}_r \text{ resolved}\}}.$$

Unresolved cases are reported separately. They are neither silently counted as covered nor silently counted as uncovered. An explicitly **empty_valid** region is different: it is numerically resolved, contains no accepted value, has analysis length zero, and is uncovered. The legacy DML schema cannot distinguish a valid empty region from other causes of missing geometry, so missing DML CR triplets remain unresolved and status-unavailable.

2.2 Monte Carlo uncertainty

For R_{resolved} resolved replications and coverage estimate $\hat{p}$, the Monte Carlo standard error is

$$\text{MCSE}(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{R_{\text{resolved}}}}.$$

At the design-cell level, $R_{\text{resolved}} \approx 500$ and $\hat{p} \approx 0.95$, giving an MCSE of about 0.98 percentage points. A rough 95% Monte Carlo interval is therefore about plus or minus two percentage points. Small deviations such as 94% versus 95% may be simulation noise in an individual cell; coverage near 84% is not.

2.3 Nonstandard confidence-region states

- A **disconnected region** contains multiple accepted blocks. It signals a nonstandard objective or weak/irregular identification, not a formatting error.
- An **empty region** rejects every candidate after a resolved evaluation. It is valid numerical output and counts as noncoverage.
- A **full-grid region** accepts every candidate in $[-1, 3]$. It covers the truth when the truth is on that grid, but offers little localization.
- An **unresolved region** has a numerical ambiguity that prevents a definitive covered/uncovered classification.

Coverage and informativeness must be read together. Wide or full-grid regions can be honest consequences of weak information; narrow regions are useful only if their calibration is credible. Test inversion and finite-sample pivotal methods motivate this set-valued perspective (Chernozhukov and Hansen 2008; Chernozhukov, Hansen, and Jansson 2009).

3 Weak Instruments and Weak Identification

An instrument must be **valid** (excluded from the structural disturbance in the relevant sense) and **relevant** (informative about endogenous treatment variation). Identification strength is not exhausted by the raw correlation between Z and D . It depends on the conditional model, the score's local sensitivity to α , nuisance adjustment, and the quantile under study. Consequently, an instrument can be informative for one structural quantile and weak for another (Jun 2008).

Conventional Wald approximations expand an estimator around a well-identified point and divide by an estimated local slope. When that slope is weak or unstable, normal approximations can be poor and standard errors can conceal the problem. Weak-identification-robust procedures instead test each candidate α and invert the test. The resulting region may be wide, disconnected, or the entire grid. Such a region is not automatically a failed estimator: it can be an honest statement that the sample does not localize the parameter (Chernozhukov and Hansen 2008).

This distinction matters directly here. DML's near-nominal coverage comes with wider regions and more full-grid outcomes. Oracle knowledge of the true support cannot remove weak identification because support knowledge creates no new instrument variation. Post-selection can look precise while under-covering if selection error or unaccounted selection uncertainty narrows the accepted set. Weak instruments are therefore part of the explanation, but not a universal explanation for all Post-selection failures.

4 Estimators

4.1 Oracle IVQR

Oracle IVQR supplies the true union of active outcome and first-stage controls to the shared Chernozhukov–Hansen inverse-IVQR routine. It is infeasible in an empirical application because the true support is unknown. Its purpose is to benchmark the cost of learning nuisance structure. Oracle performance is not a theoretical

Table 1: Validated Monte Carlo design

Component	Validated/configured value
DGPs	DGP1, DGP2, DGP3
Active support	DGP1: 5 active controls, DGP2: 10 active controls, DGP3: 5 active controls
Sample sizes	500, 1000
Control dimensions	200, 500
Instrument-strength index π	0.1, 0.25, 0.5, 1.0
Quantiles τ	0.25, 0.5, 0.75
Replications per cell	500
Design cells	144
True α by $\tau=(0.25, 0.50, 0.75)$	DGP1: (0.3255, 1.0000, 1.6745); DGP2: (0.3255, 1.0000, 1.6745); DGP3: (0.4371, 1.0000, 1.5629)
Initial α grid	[-1, 3], 21 points, spacing 0.2
CR refinement	Adaptive boundary refinement; tolerance 0.025
Nominal confidence level	95%
Critical-value multiplier	1
Estimators	DML-style residualized IVQR; true-support Oracle IVQR; mean-Lasso-union Post-selection IVQR

Note: Configured values are cross-checked against validation.json and the current simulation code.

upper bound: finite samples, grid point estimation, and weak identification remain even with the correct support.

4.2 Post-selection IVQR

The repository implementation fits mean-Lasso models for Y on X and D on X , takes the union of the selected controls, and applies the same inverse-IVQR routine. Selection is not quantile-specific, alpha-specific, cross-fitted, orthogonal, or formally selection-adjusted. It is therefore a heuristic sparse benchmark rather than an implementation of a uniformly valid post-selection theory. General post-selection results emphasize that regularization must be paired with orthogonalization or other protections for target inference (Chernozhukov, Hansen, and Spindler 2015). A short region after naive selection is not evidence that selection uncertainty has been handled.

4.3 DML-IVQR

For each candidate α , the implementation cross-fits a penalized quantile regression of $Y - D\alpha$ on X and residualizes the excluded instrument Z on X using Ridge regression. It evaluates the cross-fitted scalar score

$$\psi_i(\alpha) = [\tau - 1\{Y_i - D_i\alpha - \hat{q}_\alpha(X_i) \leq 0\}] \tilde{Z}_i$$

and inverts a weighted score statistic. Cross-fitting separates nuisance training from score evaluation, while residualization aims to reduce first-order sensitivity to nuisance error—the central DML logic (Chernozhukov et al. 2018).

The label **DML-style IVQR** is important. The code explicitly states that this is not the exact density-weighted procedure of Chen, Huang, and Tien (2021). The report therefore interprets the results as evidence for this repository’s cross-fitted, residualized implementation, not as a generic validation of every DML-IVQR estimator.

5 Monte Carlo Design

The outcome has a binary endogenous treatment generated from a latent index. DGP1 is a sparse Gaussian baseline; DGP2 is a denser sparse selection-stress design; DGP3 uses heavy-tailed, variance-normalized Student- t_5 structural shocks. The true structural effect varies with τ , so bias and coverage are evaluated against the appropriate DGP-quantile truth rather than a single constant.

Table 2: Overall finite-sample performance

Estimator	Coverage	Bias	MAE	RMSE	Estimate SD	Mean CR length	Median CR length	Full grid	Unresolved
DML-IVQR	94.90%	0.009	0.760	0.987	1.045	2.959	3.188	33.09%	0.06%
Oracle IVQR	94.19%	-0.007	0.602	0.836	0.930	2.407	2.235	19.37%	0.00%
Post-selection IVQR	92.42%	-0.057	0.624	0.859	0.928	2.380	2.199	18.22%	0.00%

Note: Coverage is conditional on resolved replications. CR denotes confidence region. Percentages use the authoritative validated summaries.

All estimators use the same design-replication keys and seeds. There are $3 \times 2 \times 2 \times 4 \times 3 = 144$ cells and 500 replications per cell for each estimator. The validated panel contains 72,000 rows per estimator and 216,000 in total, with no duplicate design-replication keys.

6 Overall Results

DML-IVQR has the highest overall coverage (94.90%), but also the longest mean region (2.959). Oracle IVQR has the smallest MAE (0.602) and RMSE (0.836), making it the strongest point-estimation benchmark. Post-selection produces a slightly shorter mean region than Oracle (2.380 versus 2.407) but covers only 92.42%. The shorter set is therefore not evidence of superior inference.

The ranking changes with the criterion. Oracle is preferable for point accuracy; DML is preferable for aggregate coverage; Post-selection is slightly more compact than Oracle but inferentially less reliable. Coverage must not be confused with precision, and precision must not be confused with validity.

7 Results by Quantile

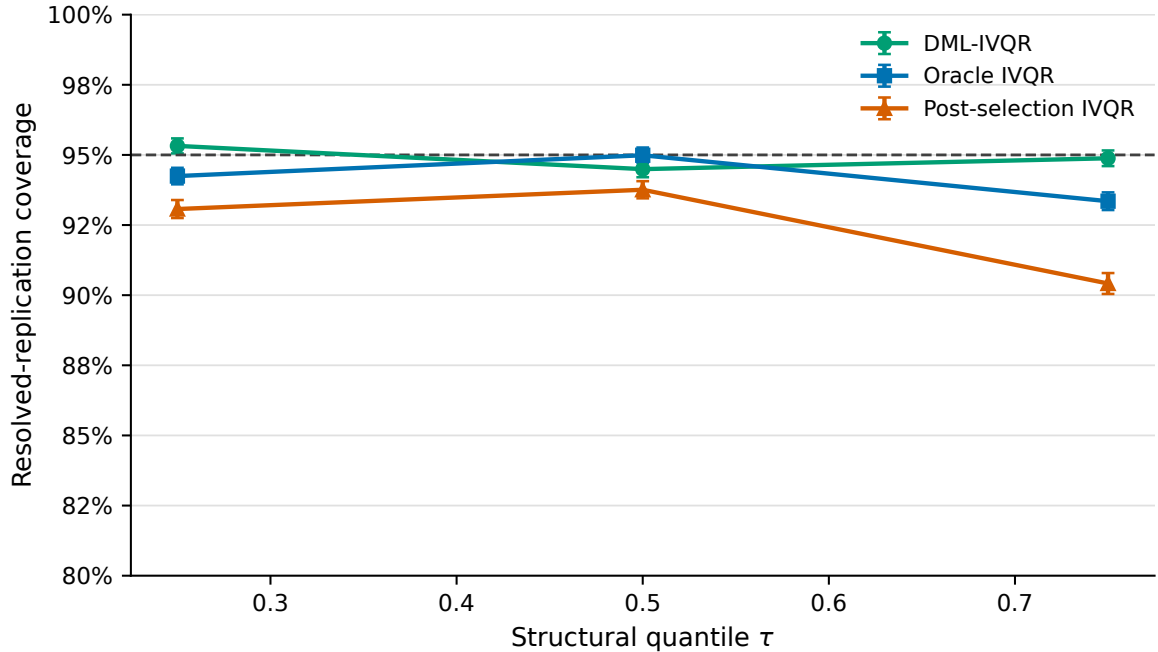


Figure 1: Resolved-replication coverage by structural quantile. Vertical bars are normal 95% Monte Carlo intervals around the aggregated coverage estimate; the dashed line is the nominal 95% target.

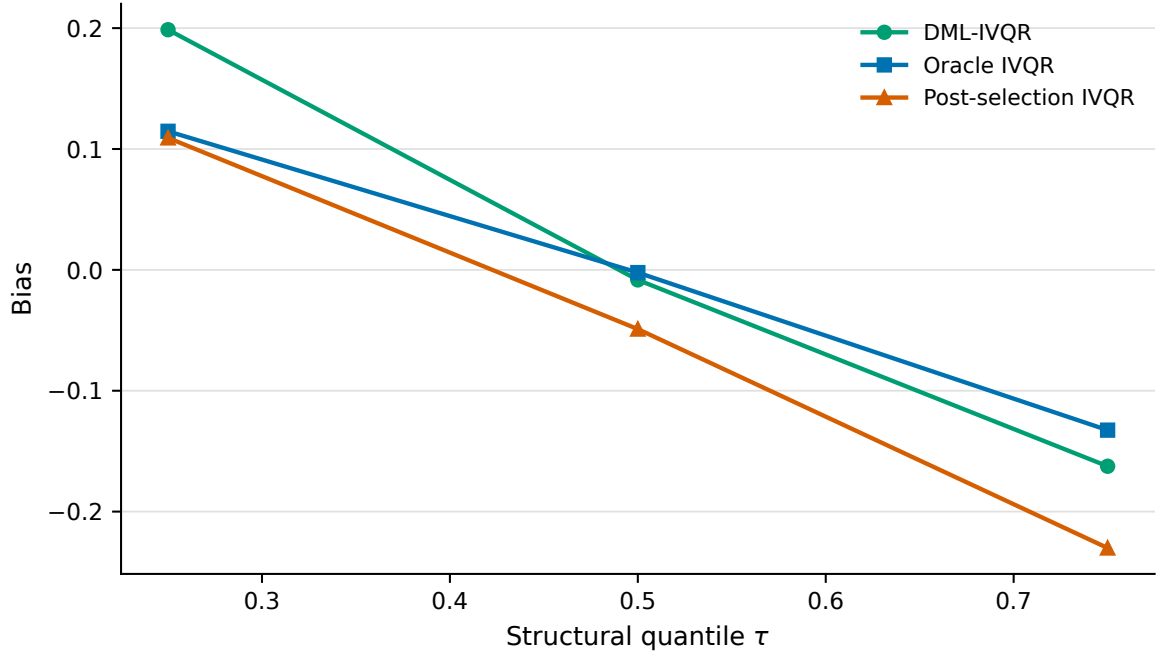


Figure 2: Bias by structural quantile. The upper-quantile Post-selection result is distinguished by a materially more negative bias.

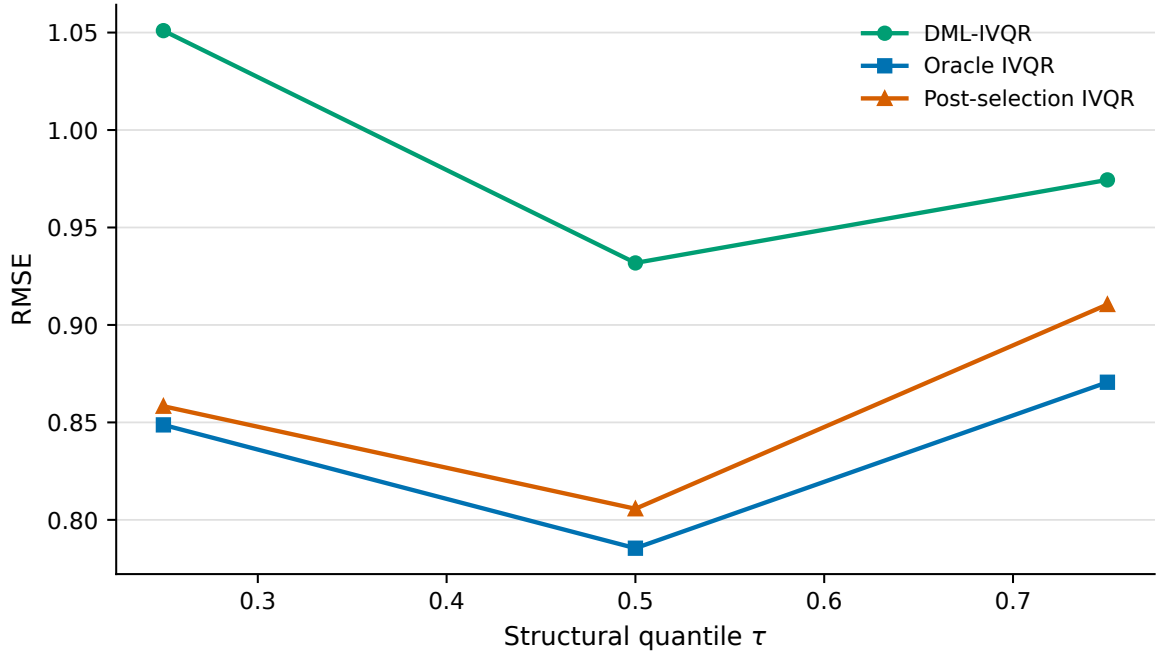


Figure 3: RMSE by structural quantile. RMSE combines variance and bias and should be read alongside coverage rather than as a substitute for it.

The main instability is concentrated in the upper quantile. Post-selection coverage is 90.42% at $\tau = 0.75$, with bias -0.230; its coverage is 93.07% at $\tau = 0.25$. Oracle also weakens at $\tau = 0.75$ but less severely, while DML remains close to nominal at each reported quantile. The coincident negative upper-quantile bias is consistent with, but does not by itself prove, a selection mechanism.

The aggregate error bars in Figure 1 are narrow because each point pools roughly 24,000 replications. They quantify Monte Carlo uncertainty, not model uncertainty. The upper-quantile Post-selection gap is far larger than that simulation uncertainty.

8 Results by Instrument Strength

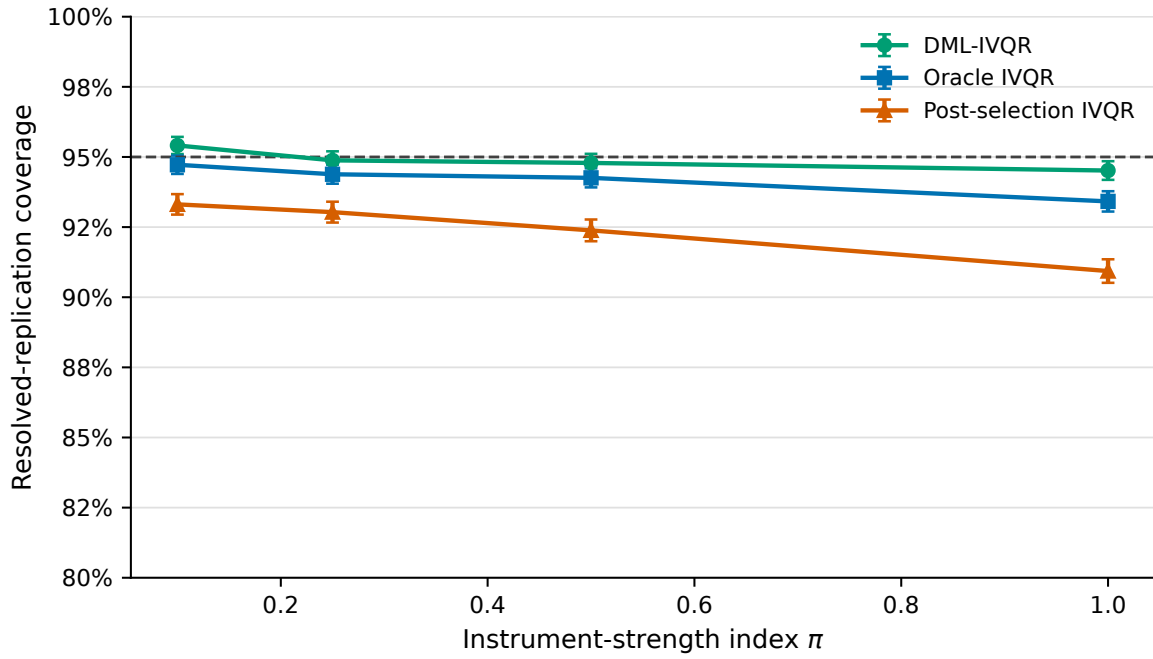


Figure 4: Coverage by the instrument-strength index. The vertical scale is explicitly restricted to 80–100% to show calibration differences; the nominal 95% line remains visible.

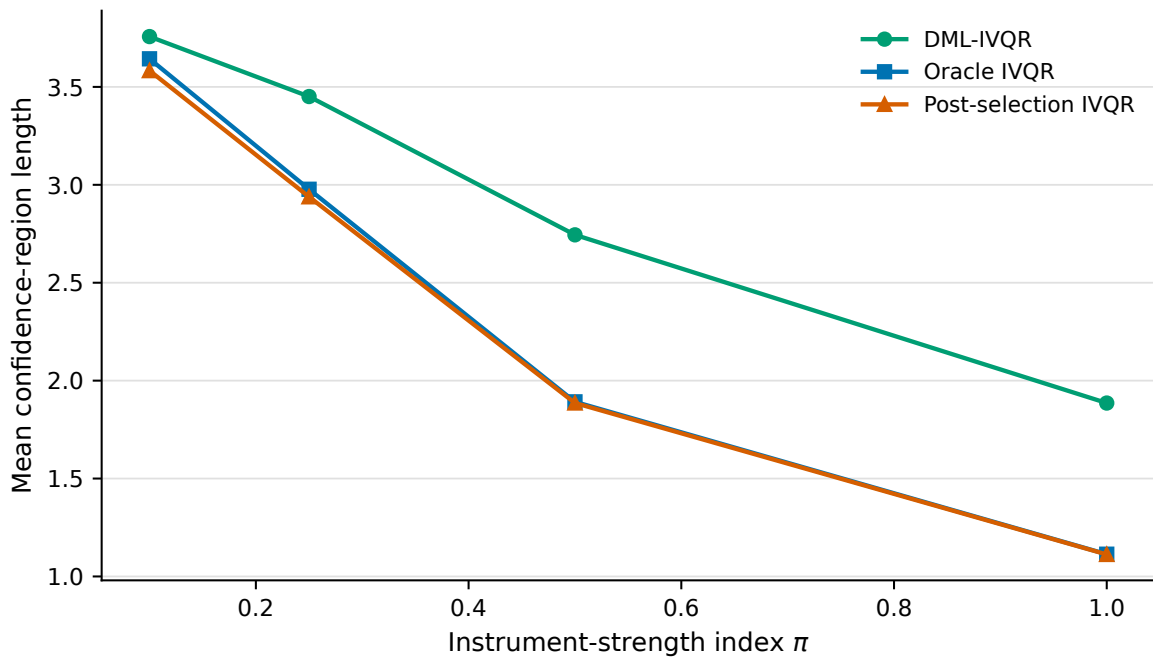


Figure 5: Mean confidence-region length by instrument strength. Falling length with stronger instruments reflects improved localization, not automatically improved calibration.

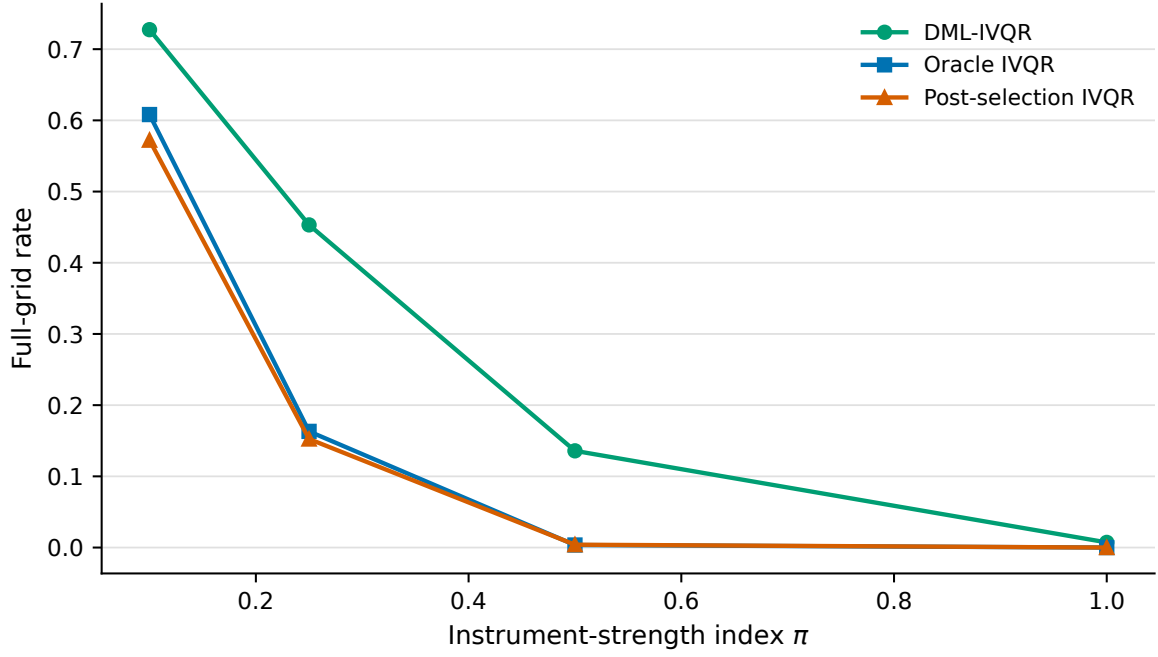


Figure 6: Full-grid rate by instrument strength. Full-grid outcomes are most common when the instrument-strength index is smallest.

All estimators become more informative as the instrument-strength index increases. For $\pi = 0.1$, mean CR lengths are 3.757 (DML), 3.643 (Oracle), and 3.583 (Post-selection). At $\pi = 1$, the corresponding lengths fall to 1.886, 1.114, and 1.112. This is the expected informativeness response to stronger identification. Post-selection undercoverage, however, persists—and is most visible at $\pi = 1$ —so weak identification is not its only plausible source.

The Oracle and Post-selection length curves are close, whereas DML is wider, especially for weak designs. Stronger instruments reduce full-grid rates sharply. Yet Post-selection coverage declines as π rises from 0.1 to 1.0. This pattern is consistent with selection-related bias becoming visible when confidence regions narrow; it is not evidence that instrument strength causes the bias.

9 Precision and Informativeness

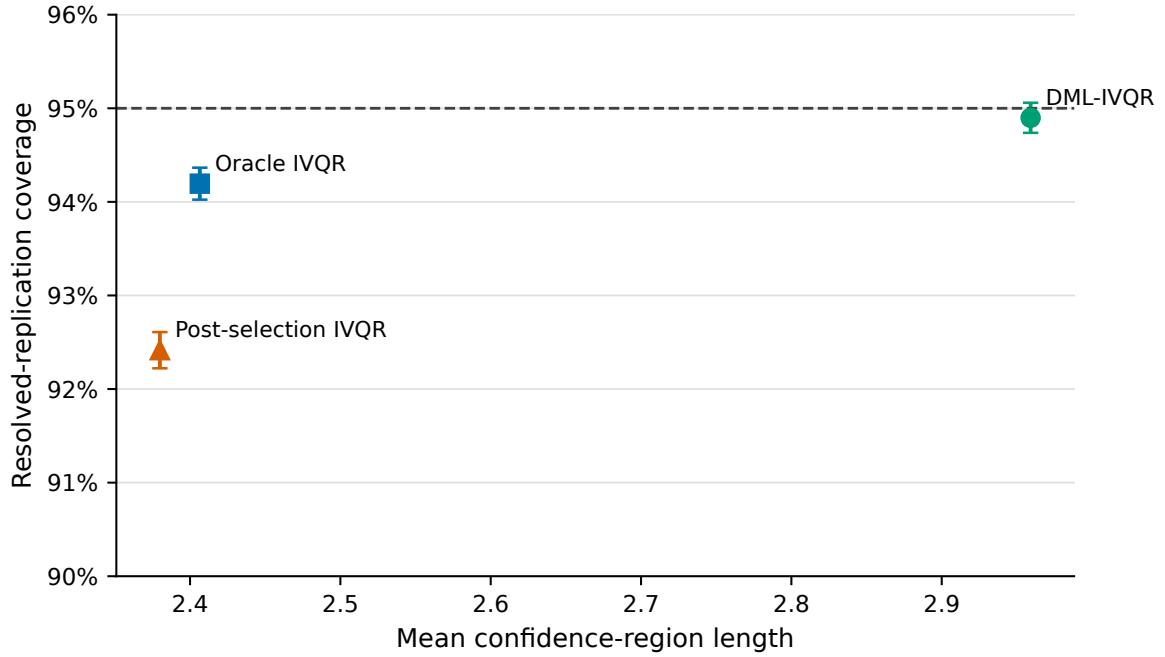


Figure 7: Overall coverage versus mean confidence-region length. Vertical bars show 95% Monte Carlo uncertainty for coverage. Moving left is attractive only if coverage remains calibrated.

The trade-off in Figure 7 is not a conventional efficiency frontier because the estimators do not all satisfy the same coverage constraint. DML occupies the high-coverage, low-informativeness position. Oracle gives the strongest balance of point accuracy and region length, though it is infeasible. Post-selection’s length is attractive in isolation but its coverage is inadequate.

Decision rule: A shorter confidence region is not preferable if it fails to achieve nominal coverage.

Full-grid DML regions should not be hidden as failures: under weak identification, they can be the correct statement that the candidate grid cannot be narrowed. They do, however, limit the estimator’s substantive usefulness. The supervisor decision must therefore balance honest uncertainty against the thesis’s need for informative empirical conclusions.

10 Worst-Performing Designs

The lowest-coverage cell is Post-selection IVQR, DGP2, $n = 500$, $p = 500$, $\pi = 1$, and $\tau = 0.75$, with coverage 83.8% and MCSE 1.65%. A 95% Monte Carlo interval is approximately [80.6%, 87.0%], far below 95%. Sampling noise from 500 replications cannot plausibly explain that gap.

The worst ten cells are all Post-selection cells and heavily concentrated at $\tau = 0.75$. Several occur with $\pi = 0.5$ or 1, so the tail failures cannot be attributed solely to the weakest-instrument configuration. They also occur across DGPs and both control dimensions. This is evidence of systematic heterogeneity in inferential performance, not one anomalous simulation cell.

11 Numerical Diagnostics

The resolved denominator excludes 43 DML rows (0.060%) and two Post-selection rows (0.003%); Oracle has no unresolved rows. DML’s legacy file contains no explicit CR status, component, iteration-warning, rank-failure, or refinement metadata. Those entries are intentionally reported as **NA**, not zero. Oracle and Post-selection record zero rank failures and zero refinement-limit hits in these validated summaries.

Table 3: Ten lowest-coverage design cells: design and coverage

Estimator	DGP	n	p	pi	tau	Resolved	Coverage	MCSE
Post-selection IVQR	DGP2	500	500	1	0.75	500	83.80%	1.65%
Post-selection IVQR	DGP3	500	500	1	0.75	500	83.80%	1.65%
Post-selection IVQR	DGP2	500	200	1	0.75	500	85.80%	1.56%
Post-selection IVQR	DGP2	500	500	0.25	0.75	500	86.80%	1.51%
Post-selection IVQR	DGP2	500	500	0.5	0.75	500	87.00%	1.50%
Post-selection IVQR	DGP2	500	200	0.5	0.75	500	88.00%	1.45%
Post-selection IVQR	DGP2	1000	500	0.5	0.75	500	88.00%	1.45%
Post-selection IVQR	DGP2	1000	500	1	0.75	500	88.00%	1.45%
Post-selection IVQR	DGP1	1000	500	1	0.75	500	88.20%	1.44%
Post-selection IVQR	DGP2	1000	200	0.1	0.75	500	88.40%	1.43%

Note: Cells are ranked by resolved-replication coverage. MCSE is the Monte Carlo standard error of coverage.

Table 4: Ten lowest-coverage design cells: performance and diagnostics

Estimator	DGP	n	p	pi	tau	Bias	RMSE	Mean CR length	Full grid	Unresolved
Post-selection IVQR	DGP2	500	500	1	0.75	-0.256	0.516	1.478	0.00%	0.00%
Post-selection IVQR	DGP3	500	500	1	0.75	-0.157	0.391	1.171	0.00%	0.00%
Post-selection IVQR	DGP2	500	200	1	0.75	-0.146	0.452	1.452	0.00%	0.00%
Post-selection IVQR	DGP2	500	500	0.25	0.75	-0.472	1.279	3.244	35.00%	0.00%
Post-selection IVQR	DGP2	500	500	0.5	0.75	-0.352	0.840	2.502	2.00%	0.00%
Post-selection IVQR	DGP2	500	200	0.5	0.75	-0.270	0.825	2.513	3.60%	0.00%
Post-selection IVQR	DGP2	1000	500	0.5	0.75	-0.154	0.570	1.836	0.00%	0.00%
Post-selection IVQR	DGP2	1000	500	1	0.75	-0.107	0.339	1.008	0.00%	0.00%
Post-selection IVQR	DGP1	1000	500	1	0.75	-0.086	0.290	0.940	0.00%	0.00%
Post-selection IVQR	DGP2	1000	200	0.1	0.75	-0.402	1.327	3.560	59.00%	0.00%

Note: Rows correspond exactly to the cells in the preceding table. The split avoids compressing fourteen columns into one table.

Table 5: Numerical and confidence-region diagnostics

Estimator	Unresolved	Empty	Disconnected	Boundary estimate	Iteration warning	Rank failure	Refinement limit
DML-IVQR	0.06%	NA	NA	7.66%	NA	NA	NA
Oracle IVQR	0.00%	0.02%	15.95%	5.35%	70.01%	0.00%	0.00%
Post-selection IVQR	0.00%	0.06%	16.48%	5.56%	89.86%	0.00%	0.00%

Note: NA means the source schema does not provide that diagnostic. It must not be interpreted as zero.

Iteration warnings are common in the richer Oracle and Post-selection schemas, but the production policy retains warning fits when their numerical output is otherwise usable. Rank-failure and refinement-limit rates are zero in these validated results. DML lacks the metadata needed to make the same statements; reporting zero would fabricate information.

11.1 Initial-grid point estimation

The point estimate $\hat{\alpha}$ is selected from the initial 21-point grid $[-1, 3]$, whose spacing is 0.2. Adaptive midpoint probing and refinement improve confidence-region topology and boundary location, but the production setting `alpha_hat_grid="initial"` does not use those added evaluations to refine the point estimate. Boundary-estimate rates are therefore reported as a diagnostic.

This discrete point-estimation rule can affect bias, RMSE, and boundary mass. It does not by itself show that inverted-region inference is invalid: point selection and test inversion use related but distinct features of the evaluated objective. Nevertheless, a refined-point robustness analysis is a sensible candidate before final thesis submission.

12 Interpretation and Thesis Decision

12.1 DML-IVQR: plausible main feasible estimator

DML is approximately nominal overall, remains close to 95% across quantiles, and has very few status-unavailable rows. Its limitation is informativeness: weak designs often generate long or full-grid regions. The appropriate thesis claim is not that DML is uniformly precise, but that it gives the most credible feasible coverage among the three implementations while revealing weak information rather than concealing it. **Provisional decision:** acceptable as the main feasible estimator, conditional on transparent reporting of region length and full-grid frequency.

12.2 Oracle IVQR: benchmark, not empirical procedure

Oracle has the best point accuracy and a useful coverage–length balance. It is mildly below nominal overall and weaker in some upper-quantile designs, confirming that nuisance support is not the only finite-sample difficulty. **Provisional decision:** retain as the infeasible benchmark for interpreting nuisance-learning costs; do not present it as an available empirical estimator.

12.3 Post-selection IVQR: comparison or redesign

Post-selection produces Oracle-like confidence-region lengths but systematic undercoverage, with a pronounced upper-quantile problem and severe worst-cell failures. Its current mean-Lasso union step does not implement formal selection-adjusted inference. **Provisional decision:** do not present it as a uniformly reliable 95% procedure. It can remain as a comparison estimator, a negative result, and evidence about the cost of imperfect variable selection. If the thesis requires it as a positive feasible method, redesign and fresh validation would be necessary; this report does not authorize or perform that simulation work.

13 Questions for the Supervisor

Decision questions

1. **DML informativeness.** Is DML-IVQR sufficiently informative for the thesis despite its wide and full-grid confidence regions, provided these are reported as evidence of weak information rather than hidden?
2. **Role of Post-selection.** Should Post-selection remain as a documented comparison and negative result, or should it be redesigned and revalidated before inclusion as a positive method?
3. **Point-estimate grid.** Should the final robustness analysis refine point estimation beyond the initial 0.2-spaced alpha grid while leaving the validated main results intact?

Table 6: Performance by sample size and control dimension

Estimator	n	p	Coverage	Bias	RMSE	Mean CR length	Full grid	Unresolved
DML-IVQR	500	200	94.72%	0.022	1.034	3.123	36.95%	0.05%
DML-IVQR	500	500	94.94%	0.022	1.073	3.238	39.96%	0.07%
DML-IVQR	1000	200	95.00%	-0.021	0.824	2.468	21.51%	0.03%
DML-IVQR	1000	500	94.94%	0.015	0.998	3.008	33.94%	0.09%
Oracle IVQR	500	200	94.17%	-0.004	0.897	2.606	23.08%	0.00%
Oracle IVQR	500	500	94.21%	-0.009	0.892	2.597	22.78%	0.00%
Oracle IVQR	1000	200	94.14%	-0.005	0.777	2.209	15.59%	0.00%
Oracle IVQR	1000	500	94.26%	-0.009	0.767	2.215	16.03%	0.00%
Post-selection IVQR	500	200	92.46%	-0.049	0.921	2.577	21.86%	0.00%
Post-selection IVQR	500	500	91.77%	-0.092	0.933	2.572	21.36%	0.00%
Post-selection IVQR	1000	200	92.86%	-0.036	0.783	2.182	14.77%	0.01%
Post-selection IVQR	1000	500	92.58%	-0.049	0.788	2.189	14.88%	0.01%

Note: Coverage uses resolved replications only.

Table 8: Detailed standardized confidence-region statuses

Estimator	Status	Rows	Resolved	Covered	Uncovered	Unresolved	Coverage
DML-IVQR	full_grid_observed_status_unavailable	23825	23825	23825	0	0	100.00%
DML-IVQR	missing_status_unavailable	43	0	0	0	43	NA
DML-IVQR	observed_status_unavailable	48132	48132	44461	3671	0	92.37%
Oracle IVQR	empty_valid	13	13	0	13	0	0.00%
Oracle IVQR	full_grid_valid	13948	13948	13948	0	0	100.00%
Oracle IVQR	valid	58039	58039	53872	4167	0	92.82%
Post-selection IVQR	empty_valid	41	41	0	41	0	0.00%
Post-selection IVQR	full_grid_valid	13115	13115	13115	0	0	100.00%
Post-selection IVQR	partially_unresolved	2	0	0	0	2	NA
Post-selection IVQR	valid	58842	58842	53422	5420	0	90.79%

Note: DML status labels explicitly state that detailed legacy status metadata are unavailable.

A Results by Sample Size and Dimension

This aggregation checks whether the headline rankings are driven by a single (n, p) configuration. It should be interpreted jointly with the complete cell table, because aggregation over DGP, π , and τ can conceal difficult subdesigns.

B Confidence-Region Status Definitions

Standardized status	Meaning	Coverage treatment
valid	Resolved nonempty region that is not the full grid	Included
full_grid_valid	Resolved region accepts the entire grid	Included
empty_valid	Resolved accepted set is empty	Included as uncovered
partially_unresolved	Some evaluations prevent definitive topology/coverage	Excluded
fully_unresolved	No definitive numerical region	Excluded
observed_status_unavailable	Legacy DML geometry observed, detailed status absent	Included using the documented complete-triplet proxy
missing_status_unavailable	Legacy DML CR geometry missing and cause unknown	Excluded and reported separately

C Selected-Control Diagnostic

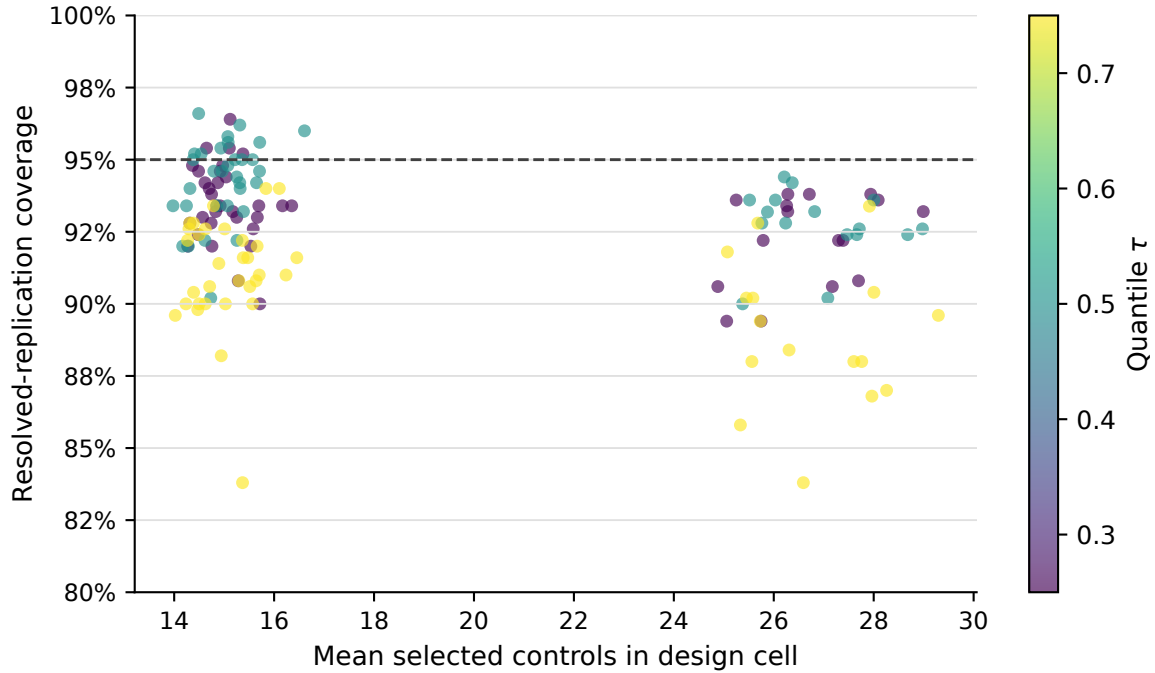


Figure 8: Post-selection design-cell coverage versus the mean number of selected controls. Color denotes the structural quantile. This descriptive plot does not identify a causal effect of selection size on coverage.

Selected-control counts are genuinely available for Post-selection and are shown only as a descriptive diagnostic. The relationship mixes DGP, sample size, dimension, instrument strength, and quantile. It cannot establish that selecting more or fewer controls causes a particular coverage outcome.

D Estimator-Specific Implementation Notes

- **Shared inversion.** Oracle and Post-selection use a Chernozhukov–Hansen-style inverse-IVQR grid routine with midpoint-assisted adaptive refinement of confidence-region boundaries.
- **Oracle support.** The active support is the exact union of nonzero outcome and first-stage coefficients: five controls in DGP1/DGP3 and ten in DGP2.
- **Post-selection.** Mean-Lasso control sets from $Y \sim X$ and $D \sim X$ are unioned. All excluded instruments are retained. The method is not quantile-specific or formally selection-adjusted.
- **DML-style procedure.** Candidate-specific penalized quantile nuisance models and Ridge instrument residualization are cross-fitted. The score is scalar because the implementation supports one excluded instrument. It is not the exact published density-weighted DML-IVQR algorithm.
- **Point versus region grid.** Point estimates use the initial grid; adaptive evaluations refine the confidence region.

E Reproducibility and Data Limitations

From the repository root, run the exact one-line Windows PowerShell command:

```
pixi run build_full_run_report
```

The task first regenerates all assets from the validated summary tables and then renders this document. The asset module does not read raw simulation files and does not recompute coverage. Its manifest explicitly records the validated summary source. To run the stages separately:

```
pixi run python -m analysis.full_run_report
quarto render documents\Full_Run_Results.qmd --to pdf
```

Paths containing parentheses must be quoted in PowerShell, although this report does not require an alternate raw Oracle path.

The principal data limitation is the legacy DML schema. Forty-three DML rows have missing CR triplets but no status field that can distinguish an empty valid region from numerical failure or another cause. DML also lacks components, disconnectedness, iteration-warning, rank-failure, and refinement-limit metadata. These fields remain **NA**. Oracle and Post-selection have richer versioned CR schemas. Finally, the current summary outputs do not prove that every historical command-line tuning label matches the file-generation environment; the report therefore describes implementation behavior and validated outputs without using old calibration tables as final evidence.

F Complete Design-Cell Results

The following long table reproduces all 432 estimator–design cells from the validated `table_05_by_design_cell.csv`. It is included for auditability rather than main-text interpretation.

Table 9: Complete results by estimator and design cell

Estimator	DGP	n	p	pi	tau	R-res.	Coverage	Bias	RMSE	Mean CR	Full grid	Unresolved
DML-IVQR	DGP1	500	200	0.1	0.25	500	94.60%	0.766	1.516	3.782	74.80%	0.00%
DML-IVQR	DGP2	500	200	0.1	0.25	500	95.80%	0.506	1.385	3.791	76.00%	0.00%
DML-IVQR	DGP3	500	200	0.1	0.25	500	95.20%	0.686	1.462	3.789	73.40%	0.00%
DML-IVQR	DGP1	500	500	0.1	0.25	500	96.20%	0.704	1.458	3.795	75.80%	0.00%
DML-IVQR	DGP2	500	500	0.1	0.25	498	96.18%	0.652	1.433	3.799	76.40%	0.40%
DML-IVQR	DGP3	500	500	0.1	0.25	498	94.78%	0.556	1.387	3.772	71.80%	0.40%
DML-IVQR	DGP1	1000	200	0.1	0.25	500	97.00%	0.462	1.342	3.684	65.40%	0.00%
DML-IVQR	DGP2	1000	200	0.1	0.25	500	96.20%	0.556	1.375	3.756	74.00%	0.00%
DML-IVQR	DGP3	1000	200	0.1	0.25	499	94.79%	0.394	1.316	3.656	59.80%	0.20%
DML-IVQR	DGP1	1000	500	0.1	0.25	498	96.39%	0.572	1.431	3.790	72.80%	0.40%
DML-IVQR	DGP2	1000	500	0.1	0.25	499	97.39%	0.602	1.404	3.793	77.00%	0.20%
DML-IVQR	DGP3	1000	500	0.1	0.25	499	95.99%	0.518	1.403	3.762	72.80%	0.20%
DML-IVQR	DGP1	500	200	0.25	0.25	500	95.80%	0.336	1.269	3.601	54.20%	0.00%
DML-IVQR	DGP2	500	200	0.25	0.25	498	94.78%	0.363	1.239	3.664	60.20%	0.40%
DML-IVQR	DGP3	500	200	0.25	0.25	500	95.40%	0.367	1.214	3.561	47.60%	0.00%
DML-IVQR	DGP1	500	500	0.25	0.25	499	95.79%	0.386	1.330	3.685	61.80%	0.20%
DML-IVQR	DGP2	500	500	0.25	0.25	500	95.40%	0.372	1.304	3.677	65.00%	0.00%
DML-IVQR	DGP3	500	500	0.25	0.25	500	96.00%	0.208	1.202	3.579	51.20%	0.00%
DML-IVQR	DGP1	1000	200	0.25	0.25	499	96.99%	0.096	0.978	3.137	20.40%	0.20%
DML-IVQR	DGP2	1000	200	0.25	0.25	499	95.59%	0.115	0.986	3.323	30.40%	0.20%
DML-IVQR	DGP3	1000	200	0.25	0.25	500	95.80%	-0.064	0.870	2.891	9.20%	0.00%
DML-IVQR	DGP1	1000	500	0.25	0.25	500	95.00%	0.317	1.208	3.533	48.60%	0.00%
DML-IVQR	DGP2	1000	500	0.25	0.25	500	94.80%	0.340	1.249	3.662	64.00%	0.00%
DML-IVQR	DGP3	1000	500	0.25	0.25	500	93.80%	0.218	1.163	3.471	43.60%	0.00%
DML-IVQR	DGP1	500	200	0.5	0.25	500	96.20%	0.054	0.925	3.055	16.80%	0.00%
DML-IVQR	DGP2	500	200	0.5	0.25	500	95.80%	0.088	0.977	3.233	27.40%	0.00%
DML-IVQR	DGP3	500	200	0.5	0.25	500	95.00%	-0.020	0.832	2.905	10.00%	0.00%
DML-IVQR	DGP1	500	500	0.5	0.25	500	96.20%	0.117	0.965	3.280	26.00%	0.00%
DML-IVQR	DGP2	500	500	0.5	0.25	500	94.60%	0.088	1.065	3.311	36.00%	0.00%
DML-IVQR	DGP3	500	500	0.5	0.25	499	95.79%	0.024	0.902	3.061	16.80%	0.20%
DML-IVQR	DGP1	1000	200	0.5	0.25	500	95.80%	-0.102	0.588	2.136	0.20%	0.00%
DML-IVQR	DGP2	1000	200	0.5	0.25	500	95.20%	-0.087	0.636	2.356	0.80%	0.00%
DML-IVQR	DGP3	1000	200	0.5	0.25	500	96.40%	-0.044	0.465	1.952	0.00%	0.00%
DML-IVQR	DGP1	1000	500	0.5	0.25	499	95.79%	0.008	0.827	2.922	10.00%	0.20%
DML-IVQR	DGP2	1000	500	0.5	0.25	499	93.39%	0.018	0.973	3.070	19.00%	0.20%
DML-IVQR	DGP3	1000	500	0.5	0.25	500	93.80%	-0.051	0.789	2.709	5.80%	0.00%
DML-IVQR	DGP1	500	200	1	0.25	500	92.60%	-0.045	0.623	2.222	0.40%	0.00%
DML-IVQR	DGP2	500	200	1	0.25	500	93.80%	-0.066	0.703	2.468	2.60%	0.00%
DML-IVQR	DGP3	500	200	1	0.25	500	95.20%	-0.060	0.561	2.106	0.00%	0.00%
DML-IVQR	DGP1	500	500	1	0.25	500	94.00%	-0.049	0.679	2.440	1.60%	0.00%
DML-IVQR	DGP2	500	500	1	0.25	500	96.20%	-0.033	0.729	2.685	4.60%	0.00%
DML-IVQR	DGP3	500	500	1	0.25	500	95.20%	-0.032	0.617	2.335	1.20%	0.00%
DML-IVQR	DGP1	1000	200	1	0.25	500	93.60%	-0.068	0.360	1.357	0.00%	0.00%
DML-IVQR	DGP2	1000	200	1	0.25	500	94.80%	-0.058	0.389	1.520	0.00%	0.00%
DML-IVQR	DGP3	1000	200	1	0.25	500	95.20%	-0.029	0.318	1.190	0.00%	0.00%
DML-IVQR	DGP1	1000	500	1	0.25	500	92.80%	-0.051	0.582	2.026	0.00%	0.00%
DML-IVQR	DGP2	1000	500	1	0.25	500	97.00%	-0.042	0.637	2.298	0.20%	0.00%
DML-IVQR	DGP3	1000	500	1	0.25	500	95.40%	-0.056	0.513	1.927	0.00%	0.00%
DML-IVQR	DGP1	500	200	0.1	0.5	497	94.57%	0.046	1.282	3.740	72.40%	0.60%
DML-IVQR	DGP2	500	200	0.1	0.5	500	95.00%	0.026	1.271	3.781	74.40%	0.00%
DML-IVQR	DGP3	500	200	0.1	0.5	499	94.79%	0.048	1.240	3.759	72.60%	0.20%
DML-IVQR	DGP1	500	500	0.1	0.5	499	94.19%	0.220	1.306	3.756	74.00%	0.20%
DML-IVQR	DGP2	500	500	0.1	0.5	500	95.20%	0.076	1.328	3.781	74.60%	0.00%
DML-IVQR	DGP3	500	500	0.1	0.5	500	95.20%	0.079	1.274	3.762	70.80%	0.00%
DML-IVQR	DGP1	1000	200	0.1	0.5	500	95.20%	-0.100	1.216	3.693	61.00%	0.00%
DML-IVQR	DGP2	1000	200	0.1	0.5	500	95.00%	0.026	1.226	3.710	67.00%	0.00%
DML-IVQR	DGP3	1000	200	0.1	0.5	500	95.20%	0.034	1.169	3.609	53.60%	0.00%

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Table 9 (continued)

Estimator	DGP	n	p	pi	tau	R-res.	Coverage	Bias	RMSE	Mean CR	Full grid	Unresolved
DML-IVQR	DGP1	1000	500	0.1	0.5	498	94.18%	0.058	1.326	3.767	71.60%	0.40%
DML-IVQR	DGP2	1000	500	0.1	0.5	500	94.40%	0.081	1.330	3.761	75.20%	0.00%
DML-IVQR	DGP3	1000	500	0.1	0.5	499	94.79%	0.078	1.258	3.703	67.60%	0.20%
DML-IVQR	DGP1	500	200	0.25	0.5	499	94.99%	-0.030	1.135	3.575	54.00%	0.20%
DML-IVQR	DGP2	500	200	0.25	0.5	500	94.60%	-0.046	1.174	3.613	52.40%	0.00%
DML-IVQR	DGP3	500	200	0.25	0.5	500	94.60%	-0.022	1.094	3.438	38.40%	0.00%
DML-IVQR	DGP1	500	500	0.25	0.5	500	94.60%	0.037	1.165	3.584	49.80%	0.00%
DML-IVQR	DGP2	500	500	0.25	0.5	499	93.99%	-0.063	1.224	3.611	54.60%	0.20%
DML-IVQR	DGP3	500	500	0.25	0.5	500	94.80%	-0.049	1.075	3.488	42.20%	0.00%
DML-IVQR	DGP1	1000	200	0.25	0.5	500	95.00%	-0.134	0.860	2.999	13.20%	0.00%
DML-IVQR	DGP2	1000	200	0.25	0.5	500	93.20%	-0.030	0.901	3.129	17.00%	0.00%
DML-IVQR	DGP3	1000	200	0.25	0.5	500	94.20%	-0.064	0.643	2.628	4.00%	0.00%
DML-IVQR	DGP1	1000	500	0.25	0.5	499	94.39%	-0.129	1.071	3.485	44.60%	0.20%
DML-IVQR	DGP2	1000	500	0.25	0.5	500	94.80%	0.080	1.188	3.518	44.20%	0.00%
DML-IVQR	DGP3	1000	500	0.25	0.5	500	96.20%	-0.016	0.974	3.320	30.80%	0.00%
DML-IVQR	DGP1	500	200	0.5	0.5	500	95.20%	0.052	0.786	2.924	10.40%	0.00%
DML-IVQR	DGP2	500	200	0.5	0.5	500	93.40%	-0.049	0.921	3.059	11.80%	0.00%
DML-IVQR	DGP3	500	200	0.5	0.5	500	93.20%	-0.033	0.651	2.529	2.60%	0.00%
DML-IVQR	DGP1	500	500	0.5	0.5	499	93.19%	-0.064	0.928	3.089	16.00%	0.20%
DML-IVQR	DGP2	500	500	0.5	0.5	500	94.40%	-0.040	0.988	3.296	24.40%	0.00%
DML-IVQR	DGP3	500	500	0.5	0.5	500	93.80%	-0.022	0.787	2.834	8.00%	0.00%
DML-IVQR	DGP1	1000	200	0.5	0.5	500	96.20%	-0.044	0.422	1.761	0.00%	0.00%
DML-IVQR	DGP2	1000	200	0.5	0.5	500	94.40%	-0.070	0.484	1.993	0.00%	0.00%
DML-IVQR	DGP3	1000	200	0.5	0.5	500	94.40%	-0.023	0.326	1.385	0.00%	0.00%
DML-IVQR	DGP1	1000	500	0.5	0.5	500	93.80%	-0.012	0.726	2.703	4.40%	0.00%
DML-IVQR	DGP2	1000	500	0.5	0.5	500	94.60%	-0.053	0.847	2.918	8.80%	0.00%
DML-IVQR	DGP3	1000	500	0.5	0.5	500	94.20%	-0.025	0.595	2.355	1.40%	0.00%
DML-IVQR	DGP1	500	200	1	0.5	500	92.80%	-0.035	0.528	1.985	0.20%	0.00%
DML-IVQR	DGP2	500	200	1	0.5	500	93.60%	0.022	0.546	2.160	0.00%	0.00%
DML-IVQR	DGP3	500	200	1	0.5	500	93.20%	0.005	0.434	1.609	0.00%	0.00%
DML-IVQR	DGP1	500	500	1	0.5	500	94.00%	-0.038	0.572	2.276	0.40%	0.00%
DML-IVQR	DGP2	500	500	1	0.5	500	93.60%	0.010	0.664	2.432	0.60%	0.00%
DML-IVQR	DGP3	500	500	1	0.5	500	94.60%	-0.036	0.470	1.879	0.00%	0.00%
DML-IVQR	DGP1	1000	200	1	0.5	500	95.20%	-0.016	0.274	1.047	0.00%	0.00%
DML-IVQR	DGP2	1000	200	1	0.5	500	95.60%	-0.055	0.292	1.143	0.00%	0.00%
DML-IVQR	DGP3	1000	200	1	0.5	500	94.00%	-0.011	0.232	0.841	0.00%	0.00%
DML-IVQR	DGP1	1000	500	1	0.5	500	94.00%	-0.019	0.442	1.759	0.00%	0.00%
DML-IVQR	DGP2	1000	500	1	0.5	500	96.00%	-0.013	0.464	1.916	0.00%	0.00%
DML-IVQR	DGP3	1000	500	1	0.5	500	95.20%	-0.035	0.367	1.429	0.00%	0.00%
DML-IVQR	DGP1	500	200	0.1	0.75	499	95.59%	-0.275	1.231	3.780	79.80%	0.20%
DML-IVQR	DGP2	500	200	0.1	0.75	500	94.00%	-0.350	1.352	3.754	77.40%	0.00%
DML-IVQR	DGP3	500	200	0.1	0.75	499	97.19%	-0.186	1.218	3.836	79.40%	0.20%
DML-IVQR	DGP1	500	500	0.1	0.75	500	93.80%	-0.302	1.269	3.760	76.80%	0.00%
DML-IVQR	DGP2	500	500	0.1	0.75	499	94.79%	-0.392	1.323	3.791	79.60%	0.20%
DML-IVQR	DGP3	500	500	0.1	0.75	500	96.00%	-0.190	1.262	3.788	76.20%	0.00%
DML-IVQR	DGP1	1000	200	0.1	0.75	500	94.60%	-0.214	1.195	3.721	70.20%	0.00%
DML-IVQR	DGP2	1000	200	0.1	0.75	499	96.19%	-0.297	1.286	3.783	77.20%	0.20%
DML-IVQR	DGP3	1000	200	0.1	0.75	499	94.99%	-0.238	1.226	3.694	67.00%	0.20%
DML-IVQR	DGP1	1000	500	0.1	0.75	500	96.00%	-0.257	1.242	3.801	78.00%	0.00%
DML-IVQR	DGP2	1000	500	0.1	0.75	499	96.79%	-0.294	1.252	3.805	79.40%	0.20%
DML-IVQR	DGP3	1000	500	0.1	0.75	500	96.60%	-0.206	1.249	3.749	72.80%	0.00%
DML-IVQR	DGP1	500	200	0.25	0.75	500	96.40%	-0.240	1.226	3.616	61.20%	0.00%
DML-IVQR	DGP2	500	200	0.25	0.75	500	95.00%	-0.265	1.213	3.664	66.40%	0.00%
DML-IVQR	DGP3	500	200	0.25	0.75	500	94.80%	-0.245	1.081	3.619	57.20%	0.00%
DML-IVQR	DGP1	500	500	0.25	0.75	500	93.00%	-0.200	1.219	3.609	65.40%	0.00%
DML-IVQR	DGP2	500	500	0.25	0.75	499	94.39%	-0.352	1.299	3.650	65.40%	0.20%
DML-IVQR	DGP3	500	500	0.25	0.75	499	95.79%	-0.188	1.183	3.635	60.20%	0.20%
DML-IVQR	DGP1	1000	200	0.25	0.75	500	94.40%	-0.112	0.901	3.209	28.00%	0.00%
DML-IVQR	DGP2	1000	200	0.25	0.75	500	94.40%	-0.201	0.948	3.394	39.00%	0.00%

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Table 9 (continued)

Estimator	DGP	n	p	pi	tau	R-res.	Coverage	Bias	RMSE	Mean CR	Full grid	Unresolved
DML-IVQR	DGP3	1000	200	0.25	0.75	500	93.40%	-0.108	0.832	2.975	15.60%	0.00%
DML-IVQR	DGP1	1000	500	0.25	0.75	498	95.18%	-0.217	1.158	3.558	56.60%	0.40%
DML-IVQR	DGP2	1000	500	0.25	0.75	498	94.18%	-0.216	1.215	3.635	64.00%	0.40%
DML-IVQR	DGP3	1000	500	0.25	0.75	499	94.19%	-0.183	1.160	3.500	51.00%	0.20%
DML-IVQR	DGP1	500	200	0.5	0.75	500	95.00%	-0.188	0.937	3.178	25.20%	0.00%
DML-IVQR	DGP2	500	200	0.5	0.75	500	93.20%	-0.193	1.032	3.230	32.00%	0.00%
DML-IVQR	DGP3	500	200	0.5	0.75	500	96.20%	-0.074	0.801	2.994	16.20%	0.00%
DML-IVQR	DGP1	500	500	0.5	0.75	499	95.99%	-0.157	0.990	3.311	38.20%	0.20%
DML-IVQR	DGP2	500	500	0.5	0.75	500	95.60%	-0.190	1.022	3.398	42.00%	0.00%
DML-IVQR	DGP3	500	500	0.5	0.75	500	94.80%	-0.090	0.919	3.096	20.20%	0.00%
DML-IVQR	DGP1	1000	200	0.5	0.75	500	93.80%	-0.031	0.483	1.943	0.20%	0.00%
DML-IVQR	DGP2	1000	200	0.5	0.75	500	95.20%	-0.082	0.550	2.248	1.20%	0.00%
DML-IVQR	DGP3	1000	200	0.5	0.75	500	95.00%	-0.050	0.416	1.690	0.00%	0.00%
DML-IVQR	DGP1	1000	500	0.5	0.75	500	91.80%	-0.081	0.839	2.929	19.40%	0.00%
DML-IVQR	DGP2	1000	500	0.5	0.75	500	95.20%	-0.148	0.867	3.165	28.40%	0.00%
DML-IVQR	DGP3	1000	500	0.5	0.75	500	95.80%	-0.124	0.725	2.788	9.40%	0.00%
DML-IVQR	DGP1	500	200	1	0.75	500	94.40%	-0.057	0.556	2.170	0.40%	0.00%
DML-IVQR	DGP2	500	200	1	0.75	500	93.20%	-0.051	0.634	2.316	1.80%	0.00%
DML-IVQR	DGP3	500	200	1	0.75	500	94.80%	-0.050	0.486	1.926	0.60%	0.00%
DML-IVQR	DGP1	500	500	1	0.75	500	95.20%	-0.052	0.621	2.470	3.00%	0.00%
DML-IVQR	DGP2	500	500	1	0.75	500	95.60%	-0.144	0.719	2.674	7.20%	0.00%
DML-IVQR	DGP3	500	500	1	0.75	500	94.80%	-0.056	0.591	2.202	0.60%	0.00%
DML-IVQR	DGP1	1000	200	1	0.75	500	94.60%	-0.024	0.288	1.109	0.00%	0.00%
DML-IVQR	DGP2	1000	200	1	0.75	500	94.60%	-0.047	0.310	1.243	0.00%	0.00%
DML-IVQR	DGP3	1000	200	1	0.75	500	93.80%	-0.037	0.247	0.952	0.00%	0.00%
DML-IVQR	DGP1	1000	500	1	0.75	500	94.20%	-0.049	0.474	1.900	0.20%	0.00%
DML-IVQR	DGP2	1000	500	1	0.75	500	93.80%	-0.064	0.537	2.183	0.20%	0.00%
DML-IVQR	DGP3	1000	500	1	0.75	500	96.00%	-0.026	0.397	1.683	0.20%	0.00%
Oracle IVQR	DGP1	500	200	0.1	0.25	500	95.20%	0.487	1.362	3.678	65.20%	0.00%
Oracle IVQR	DGP2	500	200	0.1	0.25	500	94.20%	0.495	1.379	3.663	65.60%	0.00%
Oracle IVQR	DGP3	500	200	0.1	0.25	500	95.40%	0.354	1.304	3.702	64.00%	0.00%
Oracle IVQR	DGP1	500	500	0.1	0.25	500	95.60%	0.541	1.398	3.708	64.60%	0.00%
Oracle IVQR	DGP2	500	500	0.1	0.25	500	95.60%	0.513	1.364	3.689	65.40%	0.00%
Oracle IVQR	DGP3	500	500	0.1	0.25	500	95.40%	0.277	1.244	3.627	56.40%	0.00%
Oracle IVQR	DGP1	1000	200	0.1	0.25	500	95.40%	0.442	1.274	3.610	56.20%	0.00%
Oracle IVQR	DGP2	1000	200	0.1	0.25	500	94.20%	0.467	1.325	3.615	57.20%	0.00%
Oracle IVQR	DGP3	1000	200	0.1	0.25	500	95.60%	0.248	1.210	3.558	50.80%	0.00%
Oracle IVQR	DGP1	1000	500	0.1	0.25	500	95.20%	0.334	1.248	3.616	58.40%	0.00%
Oracle IVQR	DGP2	1000	500	0.1	0.25	500	94.60%	0.377	1.268	3.618	57.60%	0.00%
Oracle IVQR	DGP3	1000	500	0.1	0.25	500	94.80%	0.233	1.174	3.552	49.20%	0.00%
Oracle IVQR	DGP1	500	200	0.25	0.25	500	95.00%	0.106	1.019	3.189	25.00%	0.00%
Oracle IVQR	DGP2	500	200	0.25	0.25	500	93.80%	0.112	1.069	3.252	26.80%	0.00%
Oracle IVQR	DGP3	500	200	0.25	0.25	500	95.40%	0.110	0.934	3.089	19.40%	0.00%
Oracle IVQR	DGP1	500	500	0.25	0.25	500	94.20%	0.104	1.021	3.179	25.60%	0.00%
Oracle IVQR	DGP2	500	500	0.25	0.25	500	94.00%	0.111	1.067	3.236	32.20%	0.00%
Oracle IVQR	DGP3	500	500	0.25	0.25	500	93.40%	0.030	0.896	2.973	14.00%	0.00%
Oracle IVQR	DGP1	1000	200	0.25	0.25	500	95.00%	0.012	0.840	2.749	6.20%	0.00%
Oracle IVQR	DGP2	1000	200	0.25	0.25	500	94.40%	0.112	0.842	2.906	11.00%	0.00%
Oracle IVQR	DGP3	1000	200	0.25	0.25	500	95.60%	-0.026	0.696	2.478	2.00%	0.00%
Oracle IVQR	DGP1	1000	500	0.25	0.25	500	95.20%	0.021	0.781	2.739	5.20%	0.00%
Oracle IVQR	DGP2	1000	500	0.25	0.25	500	95.40%	0.065	0.817	2.947	11.60%	0.00%
Oracle IVQR	DGP3	1000	500	0.25	0.25	500	93.80%	0.030	0.679	2.487	1.80%	0.00%
Oracle IVQR	DGP1	500	200	0.5	0.25	500	94.00%	-0.008	0.602	2.268	0.20%	0.00%
Oracle IVQR	DGP2	500	200	0.5	0.25	500	95.20%	0.027	0.663	2.390	1.40%	0.00%
Oracle IVQR	DGP3	500	200	0.5	0.25	500	94.80%	-0.077	0.533	2.004	0.00%	0.00%
Oracle IVQR	DGP1	500	500	0.5	0.25	500	94.80%	0.010	0.625	2.249	0.40%	0.00%
Oracle IVQR	DGP2	500	500	0.5	0.25	500	94.00%	-0.010	0.694	2.372	0.80%	0.00%
Oracle IVQR	DGP3	500	500	0.5	0.25	500	94.80%	-0.011	0.511	2.002	0.00%	0.00%
Oracle IVQR	DGP1	1000	200	0.5	0.25	500	93.80%	0.011	0.438	1.724	0.00%	0.00%

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Table 9 (continued)

Estimator	DGP	n	p	pi	tau	R-res.	Coverage	Bias	RMSE	Mean CR	Full grid	Unresolved
Oracle IVQR	DGP2	1000	200	0.5	0.25	500	92.80%	-0.007	0.487	1.828	0.00%	0.00%
Oracle IVQR	DGP3	1000	200	0.5	0.25	500	95.80%	0.008	0.377	1.463	0.00%	0.00%
Oracle IVQR	DGP1	1000	500	0.5	0.25	500	92.40%	0.000	0.486	1.700	0.00%	0.00%
Oracle IVQR	DGP2	1000	500	0.5	0.25	500	94.40%	0.020	0.485	1.840	0.00%	0.00%
Oracle IVQR	DGP3	1000	500	0.5	0.25	500	93.20%	-0.008	0.392	1.434	0.00%	0.00%
Oracle IVQR	DGP1	500	200	1	0.25	500	93.80%	-0.008	0.391	1.442	0.00%	0.00%
Oracle IVQR	DGP2	500	200	1	0.25	500	91.60%	-0.005	0.422	1.488	0.00%	0.00%
Oracle IVQR	DGP3	500	200	1	0.25	500	93.40%	-0.012	0.328	1.228	0.00%	0.00%
Oracle IVQR	DGP1	500	500	1	0.25	500	94.20%	-0.021	0.386	1.440	0.00%	0.00%
Oracle IVQR	DGP2	500	500	1	0.25	500	93.00%	0.012	0.412	1.529	0.00%	0.00%
Oracle IVQR	DGP3	500	500	1	0.25	500	93.40%	-0.013	0.332	1.216	0.00%	0.00%
Oracle IVQR	DGP1	1000	200	1	0.25	500	92.60%	-0.005	0.272	1.011	0.00%	0.00%
Oracle IVQR	DGP2	1000	200	1	0.25	500	92.60%	-0.011	0.295	1.082	0.00%	0.00%
Oracle IVQR	DGP3	1000	200	1	0.25	500	92.80%	0.017	0.226	0.838	0.00%	0.00%
Oracle IVQR	DGP1	1000	500	1	0.25	500	92.60%	0.004	0.274	1.006	0.00%	0.00%
Oracle IVQR	DGP2	1000	500	1	0.25	500	95.00%	0.013	0.272	1.068	0.00%	0.00%
Oracle IVQR	DGP3	1000	500	1	0.25	500	92.40%	0.018	0.231	0.844	0.00%	0.00%
Oracle IVQR	DGP1	500	200	0.1	0.5	500	95.20%	-0.039	1.253	3.747	67.40%	0.00%
Oracle IVQR	DGP2	500	200	0.1	0.5	500	95.60%	0.064	1.256	3.774	72.40%	0.00%
Oracle IVQR	DGP3	500	200	0.1	0.5	500	96.00%	0.018	1.246	3.702	62.60%	0.00%
Oracle IVQR	DGP1	500	500	0.1	0.5	500	94.80%	0.097	1.276	3.730	68.20%	0.00%
Oracle IVQR	DGP2	500	500	0.1	0.5	500	94.80%	0.050	1.310	3.726	68.80%	0.00%
Oracle IVQR	DGP3	500	500	0.1	0.5	500	95.60%	0.026	1.179	3.689	64.40%	0.00%
Oracle IVQR	DGP1	1000	200	0.1	0.5	500	94.40%	-0.052	1.204	3.606	54.20%	0.00%
Oracle IVQR	DGP2	1000	200	0.1	0.5	500	94.80%	0.080	1.250	3.662	60.40%	0.00%
Oracle IVQR	DGP3	1000	200	0.1	0.5	500	95.80%	0.015	1.110	3.551	47.00%	0.00%
Oracle IVQR	DGP1	1000	500	0.1	0.5	500	94.80%	0.131	1.236	3.614	57.00%	0.00%
Oracle IVQR	DGP2	1000	500	0.1	0.5	500	95.40%	-0.023	1.217	3.677	62.60%	0.00%
Oracle IVQR	DGP3	1000	500	0.1	0.5	500	95.60%	-0.048	1.071	3.573	45.40%	0.00%
Oracle IVQR	DGP1	500	200	0.25	0.5	500	94.80%	-0.058	1.013	3.270	23.00%	0.00%
Oracle IVQR	DGP2	500	200	0.25	0.5	500	94.60%	-0.065	1.055	3.352	29.00%	0.00%
Oracle IVQR	DGP3	500	200	0.25	0.5	500	97.40%	0.010	0.782	3.051	12.80%	0.00%
Oracle IVQR	DGP1	500	500	0.25	0.5	500	95.80%	0.007	0.938	3.319	22.80%	0.00%
Oracle IVQR	DGP2	500	500	0.25	0.5	500	95.40%	-0.059	1.012	3.392	34.00%	0.00%
Oracle IVQR	DGP3	500	500	0.25	0.5	500	96.00%	-0.083	0.789	3.029	13.80%	0.00%
Oracle IVQR	DGP1	1000	200	0.25	0.5	500	96.20%	-0.057	0.700	2.777	3.40%	0.00%
Oracle IVQR	DGP2	1000	200	0.25	0.5	500	93.80%	0.002	0.824	2.906	6.40%	0.00%
Oracle IVQR	DGP3	1000	200	0.25	0.5	500	95.40%	-0.018	0.543	2.356	0.40%	0.00%
Oracle IVQR	DGP1	1000	500	0.25	0.5	500	94.20%	-0.052	0.743	2.753	4.40%	0.00%
Oracle IVQR	DGP2	1000	500	0.25	0.5	500	95.80%	-0.029	0.791	2.947	8.60%	0.00%
Oracle IVQR	DGP3	1000	500	0.25	0.5	500	96.80%	-0.018	0.567	2.294	0.20%	0.00%
Oracle IVQR	DGP1	500	200	0.5	0.5	500	95.60%	-0.006	0.532	2.204	0.20%	0.00%
Oracle IVQR	DGP2	500	200	0.5	0.5	500	95.20%	-0.004	0.648	2.407	0.80%	0.00%
Oracle IVQR	DGP3	500	200	0.5	0.5	500	97.00%	0.013	0.410	1.791	0.00%	0.00%
Oracle IVQR	DGP1	500	500	0.5	0.5	500	96.20%	0.000	0.524	2.219	0.20%	0.00%
Oracle IVQR	DGP2	500	500	0.5	0.5	500	94.40%	-0.022	0.630	2.383	0.20%	0.00%
Oracle IVQR	DGP3	500	500	0.5	0.5	500	95.20%	-0.022	0.452	1.777	0.00%	0.00%
Oracle IVQR	DGP1	1000	200	0.5	0.5	500	96.60%	-0.021	0.364	1.511	0.00%	0.00%
Oracle IVQR	DGP2	1000	200	0.5	0.5	500	96.40%	-0.002	0.395	1.725	0.00%	0.00%
Oracle IVQR	DGP3	1000	200	0.5	0.5	500	94.40%	-0.008	0.276	1.175	0.00%	0.00%
Oracle IVQR	DGP1	1000	500	0.5	0.5	500	93.00%	-0.007	0.415	1.503	0.00%	0.00%
Oracle IVQR	DGP2	1000	500	0.5	0.5	500	93.80%	0.034	0.440	1.717	0.00%	0.00%
Oracle IVQR	DGP3	1000	500	0.5	0.5	500	96.20%	-0.009	0.296	1.182	0.00%	0.00%
Oracle IVQR	DGP1	500	200	1	0.5	500	94.80%	0.033	0.344	1.275	0.00%	0.00%
Oracle IVQR	DGP2	500	200	1	0.5	500	93.20%	0.034	0.373	1.367	0.00%	0.00%
Oracle IVQR	DGP3	500	200	1	0.5	500	95.00%	0.012	0.279	1.056	0.00%	0.00%
Oracle IVQR	DGP1	500	500	1	0.5	500	96.20%	-0.020	0.314	1.276	0.00%	0.00%
Oracle IVQR	DGP2	500	500	1	0.5	500	91.60%	-0.007	0.385	1.388	0.00%	0.00%
Oracle IVQR	DGP3	500	500	1	0.5	500	93.00%	-0.010	0.270	1.043	0.00%	0.00%

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Table 9 (continued)

Estimator	DGP	n	p	pi	tau	R-res.	Coverage	Bias	RMSE	Mean CR	Full grid	Unresolved
Oracle IVQR	DGP1	1000	200	1	0.5	500	93.80%	0.001	0.239	0.885	0.00%	0.00%
Oracle IVQR	DGP2	1000	200	1	0.5	500	93.20%	0.000	0.258	0.962	0.00%	0.00%
Oracle IVQR	DGP3	1000	200	1	0.5	500	92.80%	0.011	0.198	0.706	0.00%	0.00%
Oracle IVQR	DGP1	1000	500	1	0.5	500	93.00%	-0.000	0.242	0.892	0.00%	0.00%
Oracle IVQR	DGP2	1000	500	1	0.5	500	94.40%	0.005	0.243	0.956	0.00%	0.00%
Oracle IVQR	DGP3	1000	500	1	0.5	500	95.40%	-0.013	0.190	0.713	0.00%	0.00%
Oracle IVQR	DGP1	500	200	0.1	0.75	500	94.60%	-0.361	1.365	3.703	71.40%	0.00%
Oracle IVQR	DGP2	500	200	0.1	0.75	500	93.60%	-0.297	1.333	3.619	66.20%	0.00%
Oracle IVQR	DGP3	500	200	0.1	0.75	500	92.80%	-0.294	1.317	3.612	61.40%	0.00%
Oracle IVQR	DGP1	500	500	0.1	0.75	500	94.60%	-0.372	1.313	3.668	68.00%	0.00%
Oracle IVQR	DGP2	500	500	0.1	0.75	500	92.60%	-0.408	1.375	3.642	65.20%	0.00%
Oracle IVQR	DGP3	500	500	0.1	0.75	500	92.40%	-0.326	1.354	3.567	59.40%	0.00%
Oracle IVQR	DGP1	1000	200	0.1	0.75	500	96.00%	-0.385	1.315	3.633	60.80%	0.00%
Oracle IVQR	DGP2	1000	200	0.1	0.75	500	92.00%	-0.382	1.362	3.574	60.00%	0.00%
Oracle IVQR	DGP3	1000	200	0.1	0.75	500	94.20%	-0.298	1.253	3.577	54.00%	0.00%
Oracle IVQR	DGP1	1000	500	0.1	0.75	500	94.40%	-0.352	1.277	3.668	65.20%	0.00%
Oracle IVQR	DGP2	1000	500	0.1	0.75	500	95.20%	-0.429	1.377	3.697	63.80%	0.00%
Oracle IVQR	DGP3	1000	500	0.1	0.75	500	93.60%	-0.260	1.230	3.518	53.00%	0.00%
Oracle IVQR	DGP1	500	200	0.25	0.75	500	92.60%	-0.145	1.029	3.220	28.60%	0.00%
Oracle IVQR	DGP2	500	200	0.25	0.75	500	91.00%	-0.249	1.142	3.369	41.60%	0.00%
Oracle IVQR	DGP3	500	200	0.25	0.75	500	95.40%	-0.170	0.932	3.159	21.00%	0.00%
Oracle IVQR	DGP1	500	500	0.25	0.75	500	92.00%	-0.174	1.115	3.256	33.80%	0.00%
Oracle IVQR	DGP2	500	500	0.25	0.75	500	91.40%	-0.246	1.173	3.309	35.40%	0.00%
Oracle IVQR	DGP3	500	500	0.25	0.75	500	95.60%	-0.090	0.872	3.165	23.00%	0.00%
Oracle IVQR	DGP1	1000	200	0.25	0.75	500	93.40%	-0.070	0.844	2.868	10.80%	0.00%
Oracle IVQR	DGP2	1000	200	0.25	0.75	500	94.00%	-0.156	0.887	3.041	16.20%	0.00%
Oracle IVQR	DGP3	1000	200	0.25	0.75	500	91.80%	-0.058	0.751	2.572	4.40%	0.00%
Oracle IVQR	DGP1	1000	500	0.25	0.75	500	94.40%	-0.070	0.822	2.882	8.60%	0.00%
Oracle IVQR	DGP2	1000	500	0.25	0.75	500	91.60%	-0.164	0.937	3.024	19.00%	0.00%
Oracle IVQR	DGP3	1000	500	0.25	0.75	500	93.20%	-0.047	0.686	2.645	5.40%	0.00%
Oracle IVQR	DGP1	500	200	0.5	0.75	500	91.60%	-0.059	0.658	2.300	1.40%	0.00%
Oracle IVQR	DGP2	500	200	0.5	0.75	500	92.60%	-0.101	0.742	2.523	3.00%	0.00%
Oracle IVQR	DGP3	500	200	0.5	0.75	500	93.00%	-0.036	0.557	1.973	0.40%	0.00%
Oracle IVQR	DGP1	500	500	0.5	0.75	500	94.20%	-0.045	0.619	2.297	1.20%	0.00%
Oracle IVQR	DGP2	500	500	0.5	0.75	500	92.60%	-0.084	0.700	2.467	2.20%	0.00%
Oracle IVQR	DGP3	500	500	0.5	0.75	500	95.00%	-0.035	0.499	1.967	0.20%	0.00%
Oracle IVQR	DGP1	1000	200	0.5	0.75	500	92.40%	-0.015	0.460	1.643	0.00%	0.00%
Oracle IVQR	DGP2	1000	200	0.5	0.75	500	93.00%	-0.028	0.487	1.797	0.00%	0.00%
Oracle IVQR	DGP3	1000	200	0.5	0.75	500	94.40%	-0.008	0.353	1.377	0.00%	0.00%
Oracle IVQR	DGP1	1000	500	0.5	0.75	500	93.80%	-0.020	0.430	1.668	0.00%	0.00%
Oracle IVQR	DGP2	1000	500	0.5	0.75	500	93.00%	-0.019	0.490	1.848	0.20%	0.00%
Oracle IVQR	DGP3	1000	500	0.5	0.75	500	93.60%	-0.026	0.352	1.373	0.00%	0.00%
Oracle IVQR	DGP1	500	200	1	0.75	500	92.80%	-0.019	0.371	1.362	0.00%	0.00%
Oracle IVQR	DGP2	500	200	1	0.75	500	92.00%	0.001	0.398	1.433	0.00%	0.00%
Oracle IVQR	DGP3	500	200	1	0.75	500	92.00%	-0.018	0.316	1.146	0.00%	0.00%
Oracle IVQR	DGP1	500	500	1	0.75	500	93.80%	-0.005	0.331	1.347	0.00%	0.00%
Oracle IVQR	DGP2	500	500	1	0.75	500	94.60%	-0.018	0.377	1.453	0.00%	0.00%
Oracle IVQR	DGP3	500	500	1	0.75	500	91.20%	-0.016	0.322	1.150	0.00%	0.00%
Oracle IVQR	DGP1	1000	200	1	0.75	500	91.60%	0.018	0.259	0.945	0.00%	0.00%
Oracle IVQR	DGP2	1000	200	1	0.75	500	94.20%	-0.006	0.268	1.020	0.00%	0.00%
Oracle IVQR	DGP3	1000	200	1	0.75	500	94.00%	-0.015	0.219	0.792	0.00%	0.00%
Oracle IVQR	DGP1	1000	500	1	0.75	500	94.20%	-0.001	0.250	0.941	0.00%	0.00%
Oracle IVQR	DGP2	1000	500	1	0.75	500	92.80%	-0.010	0.284	1.001	0.00%	0.00%
Oracle IVQR	DGP3	1000	500	1	0.75	500	96.00%	0.009	0.209	0.788	0.00%	0.00%
Post-selection IVQR	DGP1	500	200	0.1	0.25	500	94.20%	0.549	1.404	3.634	63.60%	0.00%
Post-selection IVQR	DGP2	500	200	0.1	0.25	500	89.40%	0.482	1.423	3.525	59.40%	0.00%
Post-selection IVQR	DGP3	500	200	0.1	0.25	500	94.80%	0.429	1.332	3.658	61.00%	0.00%
Post-selection IVQR	DGP1	500	500	0.1	0.25	500	93.40%	0.532	1.394	3.632	60.60%	0.00%
Post-selection IVQR	DGP2	500	500	0.1	0.25	500	93.20%	0.425	1.328	3.545	57.60%	0.00%

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Table 9 (continued)

Estimator	DGP	n	p	pi	tau	R-res.	Coverage	Bias	RMSE	Mean CR	Full grid	Unresolved
Post-selection IVQR	DGP3	500	500	0.1	0.25	500	93.20%	0.296	1.271	3.560	52.80%	0.00%
Post-selection IVQR	DGP1	1000	200	0.1	0.25	500	96.40%	0.346	1.269	3.575	53.80%	0.00%
Post-selection IVQR	DGP2	1000	200	0.1	0.25	500	93.40%	0.471	1.335	3.556	53.00%	0.00%
Post-selection IVQR	DGP3	1000	200	0.1	0.25	500	94.60%	0.236	1.185	3.506	47.80%	0.00%
Post-selection IVQR	DGP1	1000	500	0.1	0.25	500	93.80%	0.390	1.283	3.572	52.80%	0.00%
Post-selection IVQR	DGP2	1000	500	0.1	0.25	500	93.60%	0.340	1.244	3.545	53.60%	0.00%
Post-selection IVQR	DGP3	1000	500	0.1	0.25	500	93.00%	0.289	1.173	3.502	45.40%	0.00%
Post-selection IVQR	DGP1	500	200	0.25	0.25	500	94.40%	0.094	1.000	3.146	23.60%	0.00%
Post-selection IVQR	DGP2	500	200	0.25	0.25	500	93.60%	0.204	1.088	3.167	26.40%	0.00%
Post-selection IVQR	DGP3	500	200	0.25	0.25	500	95.40%	0.107	0.955	3.032	16.60%	0.00%
Post-selection IVQR	DGP1	500	500	0.25	0.25	500	94.80%	0.096	1.009	3.192	23.40%	0.00%
Post-selection IVQR	DGP2	500	500	0.25	0.25	500	92.20%	0.071	1.115	3.130	28.20%	0.00%
Post-selection IVQR	DGP3	500	500	0.25	0.25	500	92.00%	0.000	0.956	2.878	13.00%	0.00%
Post-selection IVQR	DGP1	1000	200	0.25	0.25	500	92.80%	0.017	0.839	2.716	6.00%	0.00%
Post-selection IVQR	DGP2	1000	200	0.25	0.25	500	93.80%	0.093	0.864	2.836	9.60%	0.00%
Post-selection IVQR	DGP3	1000	200	0.25	0.25	500	95.40%	-0.026	0.669	2.435	1.80%	0.00%
Post-selection IVQR	DGP1	1000	500	0.25	0.25	500	95.20%	0.014	0.769	2.653	4.40%	0.00%
Post-selection IVQR	DGP2	1000	500	0.25	0.25	500	93.80%	0.043	0.862	2.826	8.40%	0.00%
Post-selection IVQR	DGP3	1000	500	0.25	0.25	500	93.00%	0.010	0.700	2.434	1.80%	0.00%
Post-selection IVQR	DGP1	500	200	0.5	0.25	500	94.20%	-0.026	0.617	2.219	0.20%	0.00%
Post-selection IVQR	DGP2	500	200	0.5	0.25	500	92.20%	0.038	0.697	2.314	1.20%	0.00%
Post-selection IVQR	DGP3	500	200	0.5	0.25	500	94.00%	-0.085	0.531	1.985	0.00%	0.00%
Post-selection IVQR	DGP1	500	500	0.5	0.25	500	93.40%	-0.018	0.637	2.200	0.60%	0.00%
Post-selection IVQR	DGP2	500	500	0.5	0.25	500	90.80%	-0.039	0.702	2.286	1.60%	0.00%
Post-selection IVQR	DGP3	500	500	0.5	0.25	500	93.40%	-0.054	0.524	1.934	0.00%	0.00%
Post-selection IVQR	DGP1	1000	200	0.5	0.25	500	93.40%	-0.009	0.441	1.683	0.00%	0.00%
Post-selection IVQR	DGP2	1000	200	0.5	0.25	500	93.20%	-0.001	0.492	1.762	0.00%	0.00%
Post-selection IVQR	DGP3	1000	200	0.5	0.25	500	94.60%	0.014	0.384	1.440	0.00%	0.00%
Post-selection IVQR	DGP1	1000	500	0.5	0.25	500	92.60%	0.007	0.480	1.661	0.00%	0.00%
Post-selection IVQR	DGP2	1000	500	0.5	0.25	500	93.80%	-0.008	0.493	1.805	0.00%	0.00%
Post-selection IVQR	DGP3	1000	500	0.5	0.25	500	90.00%	-0.016	0.420	1.400	0.00%	0.00%
Post-selection IVQR	DGP1	500	200	1	0.25	500	92.80%	-0.010	0.404	1.428	0.00%	0.00%
Post-selection IVQR	DGP2	500	200	1	0.25	500	89.40%	0.007	0.429	1.445	0.00%	0.00%
Post-selection IVQR	DGP3	500	200	1	0.25	500	92.00%	-0.009	0.332	1.212	0.00%	0.00%
Post-selection IVQR	DGP1	500	500	1	0.25	500	93.40%	-0.040	0.394	1.438	0.00%	0.00%
Post-selection IVQR	DGP2	500	500	1	0.25	500	90.60%	-0.002	0.434	1.463	0.00%	0.00%
Post-selection IVQR	DGP3	500	500	1	0.25	500	93.20%	-0.025	0.329	1.225	0.00%	0.00%
Post-selection IVQR	DGP1	1000	200	1	0.25	500	92.40%	-0.004	0.281	0.990	0.00%	0.00%
Post-selection IVQR	DGP2	1000	200	1	0.25	500	90.60%	0.004	0.299	1.040	0.00%	0.00%
Post-selection IVQR	DGP3	1000	200	1	0.25	500	93.00%	0.012	0.217	0.829	0.00%	0.00%
Post-selection IVQR	DGP1	1000	500	1	0.25	500	92.00%	-0.009	0.278	0.991	0.00%	0.00%
Post-selection IVQR	DGP2	1000	500	1	0.25	500	92.20%	-0.011	0.291	1.026	0.00%	0.00%
Post-selection IVQR	DGP3	1000	500	1	0.25	500	90.80%	0.013	0.248	0.846	0.00%	0.00%
Post-selection IVQR	DGP1	500	200	0.1	0.5	500	94.00%	-0.049	1.288	3.666	64.20%	0.00%
Post-selection IVQR	DGP2	500	200	0.1	0.5	500	92.80%	0.051	1.278	3.695	69.20%	0.00%
Post-selection IVQR	DGP3	500	200	0.1	0.5	500	94.80%	-0.075	1.238	3.662	59.60%	0.00%
Post-selection IVQR	DGP1	500	500	0.1	0.5	500	96.20%	0.058	1.289	3.711	69.40%	0.00%
Post-selection IVQR	DGP2	500	500	0.1	0.5	500	92.60%	0.092	1.316	3.627	63.40%	0.00%
Post-selection IVQR	DGP3	500	500	0.1	0.5	500	94.00%	0.043	1.222	3.663	60.80%	0.00%
Post-selection IVQR	DGP1	1000	200	0.1	0.5	500	94.40%	-0.071	1.192	3.559	50.00%	0.00%
Post-selection IVQR	DGP2	1000	200	0.1	0.5	499	93.19%	-0.006	1.275	3.593	55.40%	0.20%
Post-selection IVQR	DGP3	1000	200	0.1	0.5	500	95.80%	-0.017	1.112	3.500	43.60%	0.00%
Post-selection IVQR	DGP1	1000	500	0.1	0.5	500	94.20%	0.108	1.242	3.605	56.00%	0.00%
Post-selection IVQR	DGP2	1000	500	0.1	0.5	500	92.40%	0.016	1.254	3.580	55.20%	0.00%
Post-selection IVQR	DGP3	1000	500	0.1	0.5	500	96.00%	-0.044	1.077	3.521	44.20%	0.00%
Post-selection IVQR	DGP1	500	200	0.25	0.5	500	93.40%	-0.100	1.008	3.255	22.80%	0.00%
Post-selection IVQR	DGP2	500	200	0.25	0.5	500	92.80%	-0.105	1.101	3.264	26.00%	0.00%
Post-selection IVQR	DGP3	500	200	0.25	0.5	500	95.40%	-0.056	0.834	3.032	13.20%	0.00%
Post-selection IVQR	DGP1	500	500	0.25	0.5	500	94.60%	-0.078	0.975	3.258	21.80%	0.00%

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Table 9 (continued)

Estimator	DGP	n	p	pi	tau	R-res.	Coverage	Bias	RMSE	Mean CR	Full grid	Unresolved
Post-selection IVQR	DGP2	500	500	0.25	0.5	500	93.60%	-0.090	1.066	3.315	32.60%	0.00%
Post-selection IVQR	DGP3	500	500	0.25	0.5	500	94.60%	-0.198	0.861	3.009	11.60%	0.00%
Post-selection IVQR	DGP1	1000	200	0.25	0.5	500	94.60%	-0.115	0.730	2.752	4.40%	0.00%
Post-selection IVQR	DGP2	1000	200	0.25	0.5	500	93.60%	-0.048	0.841	2.880	7.40%	0.00%
Post-selection IVQR	DGP3	1000	200	0.25	0.5	500	95.60%	-0.081	0.552	2.347	1.60%	0.00%
Post-selection IVQR	DGP1	1000	500	0.25	0.5	500	95.00%	-0.056	0.752	2.747	5.20%	0.00%
Post-selection IVQR	DGP2	1000	500	0.25	0.5	500	93.20%	-0.094	0.813	2.891	7.80%	0.00%
Post-selection IVQR	DGP3	1000	500	0.25	0.5	500	95.60%	-0.032	0.587	2.321	0.60%	0.00%
Post-selection IVQR	DGP1	500	200	0.5	0.5	500	95.20%	-0.045	0.559	2.203	0.20%	0.00%
Post-selection IVQR	DGP2	500	200	0.5	0.5	500	94.40%	-0.088	0.664	2.348	0.80%	0.00%
Post-selection IVQR	DGP3	500	200	0.5	0.5	500	96.60%	-0.045	0.431	1.833	0.00%	0.00%
Post-selection IVQR	DGP1	500	500	0.5	0.5	500	95.00%	-0.108	0.617	2.225	0.00%	0.00%
Post-selection IVQR	DGP2	500	500	0.5	0.5	500	92.40%	-0.177	0.697	2.344	1.00%	0.00%
Post-selection IVQR	DGP3	500	500	0.5	0.5	500	95.00%	-0.094	0.480	1.815	0.00%	0.00%
Post-selection IVQR	DGP1	1000	200	0.5	0.5	500	95.00%	-0.055	0.367	1.509	0.00%	0.00%
Post-selection IVQR	DGP2	1000	200	0.5	0.5	500	94.20%	-0.039	0.438	1.678	0.00%	0.00%
Post-selection IVQR	DGP3	1000	200	0.5	0.5	500	95.20%	-0.031	0.283	1.190	0.00%	0.00%
Post-selection IVQR	DGP1	1000	500	0.5	0.5	500	92.20%	-0.067	0.427	1.529	0.00%	0.00%
Post-selection IVQR	DGP2	1000	500	0.5	0.5	500	92.60%	-0.042	0.459	1.698	0.00%	0.00%
Post-selection IVQR	DGP3	1000	500	0.5	0.5	500	94.20%	-0.042	0.314	1.208	0.00%	0.00%
Post-selection IVQR	DGP1	500	200	1	0.5	500	92.20%	-0.033	0.361	1.281	0.00%	0.00%
Post-selection IVQR	DGP2	500	200	1	0.5	500	90.00%	-0.025	0.383	1.343	0.00%	0.00%
Post-selection IVQR	DGP3	500	200	1	0.5	500	93.40%	-0.032	0.285	1.065	0.00%	0.00%
Post-selection IVQR	DGP1	500	500	1	0.5	500	93.40%	-0.103	0.362	1.306	0.00%	0.00%
Post-selection IVQR	DGP2	500	500	1	0.5	500	90.20%	-0.104	0.414	1.404	0.00%	0.00%
Post-selection IVQR	DGP3	500	500	1	0.5	500	90.20%	-0.070	0.287	1.073	0.00%	0.00%
Post-selection IVQR	DGP1	1000	200	1	0.5	500	92.00%	-0.020	0.247	0.895	0.00%	0.00%
Post-selection IVQR	DGP2	1000	200	1	0.5	500	93.60%	-0.036	0.261	0.947	0.00%	0.00%
Post-selection IVQR	DGP3	1000	200	1	0.5	500	92.00%	-0.010	0.208	0.706	0.00%	0.00%
Post-selection IVQR	DGP1	1000	500	1	0.5	500	93.40%	-0.042	0.251	0.891	0.00%	0.00%
Post-selection IVQR	DGP2	1000	500	1	0.5	500	92.40%	-0.051	0.257	0.951	0.00%	0.00%
Post-selection IVQR	DGP3	1000	500	1	0.5	500	93.20%	-0.046	0.205	0.714	0.00%	0.00%
Post-selection IVQR	DGP1	500	200	0.1	0.75	500	92.40%	-0.348	1.336	3.660	68.80%	0.00%
Post-selection IVQR	DGP2	500	200	0.1	0.75	500	92.80%	-0.357	1.334	3.575	63.40%	0.00%
Post-selection IVQR	DGP3	500	200	0.1	0.75	500	92.80%	-0.335	1.311	3.597	60.20%	0.00%
Post-selection IVQR	DGP1	500	500	0.1	0.75	500	91.60%	-0.434	1.347	3.595	62.60%	0.00%
Post-selection IVQR	DGP2	500	500	0.1	0.75	500	89.60%	-0.522	1.429	3.500	59.00%	0.00%
Post-selection IVQR	DGP3	500	500	0.1	0.75	500	91.60%	-0.364	1.367	3.513	57.20%	0.00%
Post-selection IVQR	DGP1	1000	200	0.1	0.75	500	92.60%	-0.340	1.310	3.575	58.20%	0.00%
Post-selection IVQR	DGP2	1000	200	0.1	0.75	500	88.40%	-0.402	1.327	3.560	59.00%	0.00%
Post-selection IVQR	DGP3	1000	200	0.1	0.75	500	92.60%	-0.260	1.241	3.520	49.40%	0.00%
Post-selection IVQR	DGP1	1000	500	0.1	0.75	500	94.00%	-0.388	1.327	3.622	60.40%	0.00%
Post-selection IVQR	DGP2	1000	500	0.1	0.75	499	93.39%	-0.473	1.387	3.591	59.00%	0.20%
Post-selection IVQR	DGP3	1000	500	0.1	0.75	500	94.00%	-0.254	1.218	3.487	50.20%	0.00%
Post-selection IVQR	DGP1	500	200	0.25	0.75	500	92.80%	-0.306	1.109	3.250	27.40%	0.00%
Post-selection IVQR	DGP2	500	200	0.25	0.75	500	90.20%	-0.320	1.205	3.265	34.20%	0.00%
Post-selection IVQR	DGP3	500	200	0.25	0.75	500	92.20%	-0.267	0.980	3.091	19.60%	0.00%
Post-selection IVQR	DGP1	500	500	0.25	0.75	500	90.00%	-0.335	1.171	3.274	29.80%	0.00%
Post-selection IVQR	DGP2	500	500	0.25	0.75	500	86.80%	-0.472	1.279	3.244	35.00%	0.00%
Post-selection IVQR	DGP3	500	500	0.25	0.75	500	91.00%	-0.219	1.012	3.198	23.40%	0.00%
Post-selection IVQR	DGP1	1000	200	0.25	0.75	500	91.40%	-0.131	0.898	2.850	9.80%	0.00%
Post-selection IVQR	DGP2	1000	200	0.25	0.75	500	91.80%	-0.254	0.952	3.010	17.20%	0.00%
Post-selection IVQR	DGP3	1000	200	0.25	0.75	500	90.00%	-0.091	0.765	2.570	3.60%	0.00%
Post-selection IVQR	DGP1	1000	500	0.25	0.75	500	92.00%	-0.181	0.870	2.858	10.20%	0.00%
Post-selection IVQR	DGP2	1000	500	0.25	0.75	500	90.40%	-0.296	1.042	2.992	14.60%	0.00%
Post-selection IVQR	DGP3	1000	500	0.25	0.75	500	92.20%	-0.185	0.775	2.679	5.80%	0.00%
Post-selection IVQR	DGP1	500	200	0.5	0.75	500	89.60%	-0.168	0.717	2.384	1.40%	0.00%
Post-selection IVQR	DGP2	500	200	0.5	0.75	500	88.00%	-0.270	0.825	2.513	3.60%	0.00%
Post-selection IVQR	DGP3	500	200	0.5	0.75	500	90.00%	-0.166	0.641	2.023	0.20%	0.00%

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Table 9 (continued)

Estimator	DGP	n	p	pi	tau	R-res.	Coverage	Bias	RMSE	Mean CR	Full grid	Unresolved
Post-selection IVQR	DGP1	500	500	0.5	0.75	500	90.60%	-0.231	0.726	2.392	1.40%	0.00%
Post-selection IVQR	DGP2	500	500	0.5	0.75	500	87.00%	-0.352	0.840	2.502	2.00%	0.00%
Post-selection IVQR	DGP3	500	500	0.5	0.75	500	91.60%	-0.203	0.620	2.064	0.20%	0.00%
Post-selection IVQR	DGP1	1000	200	0.5	0.75	500	89.80%	-0.093	0.483	1.658	0.00%	0.00%
Post-selection IVQR	DGP2	1000	200	0.5	0.75	500	89.40%	-0.102	0.529	1.779	0.00%	0.00%
Post-selection IVQR	DGP3	1000	200	0.5	0.75	500	90.60%	-0.070	0.381	1.374	0.00%	0.00%
Post-selection IVQR	DGP1	1000	500	0.5	0.75	500	92.60%	-0.114	0.470	1.689	0.00%	0.00%
Post-selection IVQR	DGP2	1000	500	0.5	0.75	500	88.00%	-0.154	0.570	1.836	0.00%	0.00%
Post-selection IVQR	DGP3	1000	500	0.5	0.75	500	91.00%	-0.132	0.404	1.406	0.00%	0.00%
Post-selection IVQR	DGP1	500	200	1	0.75	500	90.40%	-0.152	0.423	1.376	0.00%	0.00%
Post-selection IVQR	DGP2	500	200	1	0.75	500	85.80%	-0.146	0.452	1.452	0.00%	0.00%
Post-selection IVQR	DGP3	500	200	1	0.75	500	90.00%	-0.114	0.365	1.172	0.00%	0.00%
Post-selection IVQR	DGP1	500	500	1	0.75	500	90.80%	-0.190	0.437	1.420	0.00%	0.00%
Post-selection IVQR	DGP2	500	500	1	0.75	500	83.80%	-0.256	0.516	1.478	0.00%	0.00%
Post-selection IVQR	DGP3	500	500	1	0.75	500	83.80%	-0.157	0.391	1.171	0.00%	0.00%
Post-selection IVQR	DGP1	1000	200	1	0.75	500	90.00%	-0.038	0.273	0.934	0.00%	0.00%
Post-selection IVQR	DGP2	1000	200	1	0.75	500	90.20%	-0.078	0.295	0.995	0.00%	0.00%
Post-selection IVQR	DGP3	1000	200	1	0.75	500	93.40%	-0.066	0.233	0.795	0.00%	0.00%
Post-selection IVQR	DGP1	1000	500	1	0.75	500	88.20%	-0.086	0.290	0.940	0.00%	0.00%
Post-selection IVQR	DGP2	1000	500	1	0.75	500	88.00%	-0.107	0.339	1.008	0.00%	0.00%
Post-selection IVQR	DGP3	1000	500	1	0.75	500	90.80%	-0.064	0.231	0.797	0.00%	0.00%

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